



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

COMMON SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

I/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

GROUP – A (CIVIL, CSE, IT) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1104	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1106	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1108	History of Science and Technology	2	2	--	--	2	30	70	100
B16 ENG 1110	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1112	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1114	Sports (Audit)	--	--	--	--	3	--	--	--
Total		26	17	5	6	31	280	520	800

Note: Add-On Course: Technical English *

*** Common to both Group - A and Group - B**

Code: B16 ENG 1101

ENGLISH
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zuckerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives, Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period $2C$ (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler's Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor's Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange's Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz's Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli's Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton's Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy's linear equation – Legendre's Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler's Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval's Formula.

TEXT BOOK:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

MATHEMATICS – II
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODEs by the use of Laplace transforms.
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Also utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouché's Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by 't' – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

CHEMISTRY
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinforced plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS**Water Chemistry**

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement – Cement concrete -RCC –Decay of concrete.
Refractories: Classification-Properties and Engineering applications.
Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry

Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings- Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics

Definition-Types of polymerization – mechanism of free radical polymerization. plastics- Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants

Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasoline-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.

LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15th edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction to C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if...else statement, Nesting of if...else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops ,One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointers, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

HISTORY OF SCIENCE AND TECHNOLOGY

(Common to CIVIL, CSE & IT)

Theory : 2 periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS

Historical Perspective of Science and Technology

Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence

Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas

Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** – Effects of a nuclear explosion and India's safety measures.

Impact of Science and Technology in Major Areas

Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques- Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques; **Appropriate Technology** - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East – West Press (P) Ltd., New Delhi.

REFERENCE BOOK:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

CHEMISTRY LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl(Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^{H} Determination by using P^{H} meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stored in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that you are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{\text{rd}}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

SPORTS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD-ON COURSE
TECHNICAL ENGLISH
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.
2. Writing Skills
Business letters, E-mails
Report Writing
Writing Proposals & Business Quotations
Brochures, Newsletters
How to write an abstract & research paper
3. Speaking Skills
Technical Descriptions
PPT Presentations
Paper Presentations
4. Vocabulary
Technical Terms
5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

GROUP – A (CIVIL, CSE & IT) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1202	Physics	4	3	1	--	4	30	70	100
B16 ENG 1204	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1206	Professional Ethics & Moral values	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1209	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1211	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	NCC/NSS (Audit)	--	--	--	--	3	--	--	--
Total		24	13	3	12	31	300	500	800

Note: Add-On Course: Technology Course-I *

*** Common to both Group-A and Group-B**

Civil: B16CE1208: Building Materials and Building Construction

CSE: B16CS1208: Probability, Statistics & Queueing Theory

IT: B16CS1208: Probability, Statistics & Queueing Theory

Code: B16 ENG 1201

MATHEMATICS – III
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry

Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1

Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2

Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms

Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval's identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

PHYSICS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical Theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS**Thermodynamics**

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson-Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS:

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India (pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS:

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

ENGINEERING GRAPHICS
(Common to CIVIL, CSE & IT)

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS

Introduction

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

PROFESSIONAL ETHICS AND MORAL VALUES
(Common to CIVIL, CSE & IT)

Theory : 2 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**Ethics and Human Values**

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

BUILDING MATERIALS AND BUILDING CONSTRUCTION
(For Civil Engineering)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To teach students about the Physical and Mechanical properties of construction materials and their respective testing procedure.
2. To teach students about the building materials available in market to be used for many components of building industry.
3. To teach students about the principles and methods to be followed in constructing various components of a building.
4. To teach students about the deterioration and repair of buildings.

COURSE OUTCOMES:

1. Learn and identify the relevant physical and mechanical properties pertaining to the construction industry.
2. Demonstrate the relevant BIS testing procedure to be carried out to ascertain the quality of building materials.
3. Develop ability to choose the modern construction material appropriate to the climate and functional aspects of the buildings.
4. Ability to supervise the construction technique to be followed in brick, stone and hollow block masonry, concreting, flooring, roofing, plastering and painting etc.
5. Learn about the causes of deterioration, crack pattern, and assessment of damages.
6. Learn about the construction techniques in repairing of buildings.

SYLLABUS**Bricks and Clay Products**

Bricks: Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests, special forms of Bricks and their uses. Clay Products: Various types of tile manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, porcelain, glazing of tiles etc.

Wood, Wood Based Products

Wood: cross section details of trees, their general properties, various types of defects in wood and timber, Methods of seasoning and their importance, various Mechanical Properties of timber, Decay of timber, Wood based Products: Veneers, Plywood and its types, Manufacturing of Plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block Boards, Batten board, Hard board, Particle boards and Composite boards.

Paints, Varnishes

Paints and Varnishes: Constituents and characteristics of paints, types of paint, their uses and preparation on different surfaces, painting defects, causes and remedies. Constituents of varnishes, uses of varnishes, different kinds of varnish, polishes. Painting of interior walls, exterior walls, wooden doors and windows – steel windows – various types of paints (chemistry of paints not included) including distempers; emulsion paints etc., Varnishes wood work finishing types. **Plastic:** definition, classification of plastic and modern

development of plastic, chemistry of plastic, polymerization, manufacturing process and their uses. **Glass:** properties of glass, types of glass and manufacturing. **Asbestos & Asphalt Bitumen & Tar :** Availability and uses of asbestos, properties of asbestos, various types of asbestos, difference between asphalt & bitumen, Types, uses and properties of Asphalt & Bitumen, composition of coal tar, wood tar, mineral tar and Naphtha.

Cements

Natural and artificial cements, types of various artificial cements and their uses. Wet and dry process of manufacturing ordinary Portland cement (OPC), Chemical and Physical analysis of OPC, various field and Laboratory Tests on OPC as per IS code. Storing of cement in the field and godowns

Foundations

Types of Foundations : Strip, Isolated, Strap, Combined Footings, Raft – Mat –Slab and Beam Raft, Box Type Raft, inverted arch foundations, SHELL foundations, Grillage foundations – Minimum depth of Foundation – Bearing capacity of soils **Masonry:** Different types of Stone Masonry – Plan, Elevation, Sections of Stone Masonry Works – Brick Masonry – Different Types of Bonds – Plan, elevation and Section of Brick Bonds upto Two-Brick wall thickness – Partition walls – Different types of Block Masonry – Hollow concrete Blocks – FAL- G Blocks, Hollow Clay Blocks.

Roofing

Mangalore tiled Roof, RCC roof, Madras Terrace, Hollow Tiled Roof, Asbestos Cement, Fibre glass, Aluminium G.I. Sheet roofing's. **Form Work, Scaffolding:** form work, Types of formwork, centering-scaffolding-types of scaffolding. **Trusses:** King Post & Queen Post Trusses – Steel roof Truss for 12m Span with details. **Wooden Doors And Windows:** Parallel – Glazed – Flush shutters, Plywood, Particle Board Shutters – Aluminum, PVC, Steel doors, windows and ventilators, various types of windows, Glazing – different varieties. **Stair Cases:** Stair cases or Stairway design (Architectural design or planning only) various types such as, straight flight – dog legged, quarter landing, open spiral, spiral stairs etc.

Concrete Technology and Mix Design

Polymer Concrete, Types of cement concretes, Ingredients and their characteristics, Cement concrete properties and relevant tests, storage, batching, mixing & transporting, placing & vibrating and curing. Concrete grades & mix designs upto M 20 as per IS code. Introduction to polymer concrete and it's uses.

TEXT BOOKS:

1. "Engineering Materials" [Material Science], by Rangwala 'Charotar Publications'
2. "Building Materials" by S.K. Duggal New Age International Publishers.
3. "Building Construction" by B.C. Punmia. Laxmi Publications.
4. "Concrete Technology Theory & Practice" by M.S.Shetty, 'S.Chand & Company Ltd.'
5. "Engineering Materials", 5th edition, By Surindra Singh,, Konark Publishers Pvt. Ltd., New Delhi.

REFERENCE BOOKS:

1. "Civil Engg. Materials", by Technical Teachers' Training Institute, Chandigarh, Tata-McGraw-Hill Publishing Company Ltd., New Delhi.
2. "Materials of construction", by R.C. Smith, McGraw-Hill Company, New York.
3. "Materials of construction", by D.N. Ghose, Tata-McGraw-Hill.
4. "Engineering Materials", By Sushil Kumar, Metropolitan Book Co., Pvt. Ltd, New Delhi
5. "Building Construction" Vol.II & III By W.B. McKay, E.L.B.S. and Longman, UK.
6. "Construction Technology" by R. Chudly Vols I & II 2nd Edition Longman, UK.

PROBABILITY, STATISTICS AND QUEUING THEORY
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorials	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To discuss various definitions of probability and related theorems, understand the concepts of prior and posterior probabilities used in Baye's theorem and their use in decision making.
2. To illustrate the concept of a random variable and their properties
3. To learn the assumptions, p.m.f.'s /p.d.f.'s and analyse various statistical measures of a few discrete/continuous distributions.
4. To understand and compute the correlation coefficient, and estimation techniques from regression lines.
5. To develop a framework for testing of hypothesis in giving inferences about Population Parameter.
6. To study Queuing models and their Characteristics.

COURSE OUTCOMES:

At the end of the course a student able to

1. Handle the situation of uncertainty in decision making in our day-to-day life.
2. Identify the random variable as discrete/continuous and analyse it.
3. Predict the distribution suitable for the given data from its moments.
4. Measure the intensity of association between the variables and to fit a best suitable Curve for the given data.
5. Decide the test applicable for giving inference about Population Parameter based on Sample statistic.
6. Make business decisions about the resources needed to provide a service in day-to-day life applications including telecommunication, traffic engineering, computing and the design of factories, shops, offices and hospitals.

SYLLABUS**Probability:**

Various definitions of probability, Addition theorem, Conditional probability, Multiplication theorem, Baye's theorem of probability and Geometric probability.

Random Variables And Their Properties:

Definition, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function. Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation (**without proofs**), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations, Moment generating Function, Characteristic Function and Probability generating Function of a Random Variable.

Distributions:

Discrete Distributions:

Binomial distribution and Poisson distribution (Definition, Mean, Variance, m.g.f., Characteristic function, p.g.f., Fitting of distributions)

Continuous Distributions: Uniform distribution, its Mean and Variance. Normal Distribution and its Properties (without proofs), Standard Normal Variate, Importance of Normal distribution. Exponential Distribution, Definition, Mean Variance Memory less property of Exponential distribution

Bivariate Analysis and Curve Fitting:

CORRELATION: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient.

CURVE FITTING, Method of least Squares, Fitting of a Straight line, Fitting of a Parabola

REGRESSION ANALYSIS: Regression Lines, Regression Coefficients and their properties (without proofs)

Sampling Theory:

Sample, population, statistic, parameter, Sampling distribution, standard error, interval estimation. Testing of Hypothesis: Formulation of Null hypothesis, Alternative hypothesis, Critical region, Critical value, level of significance, Statistical Inference, Type-I-error, Type-II-error, One-tailed/Two-tailed test.

Large Sample Theory

Test of significance of single sample proportion, Test of significance for difference of proportions, Test of significance of single sample mean, Test of significance for difference of means.

Small Sample Theory: Degrees of freedom, Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means.

F-distribution: definition, F-test for equality of two population variances. Chi-square distribution definition, Chi-square test for goodness of fit, Chi-square test for single Population Variance.

Queuing Theory: Queue description, Birth and Death Process, Distribution of Inter-arrival times, Distribution of service times, Kendall's representation of a queueing model, Operating characteristics of a queueing model, steady-state solutions of $\{M/M/1: \infty/FCFS\}$ Model and $\{M/M/1; N/FCFS\}$ Model.

TEXT BOOK:

1. Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.

REFERENCE BOOKS:

1. Probability & Statistics with Reliability, Queueing and Computer Applications by Kishore.S.Trivedi, Prentice Hall of India, 1999.
2. Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers.

PHYSICS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde’s Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton’s Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy’s constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee’s apparatus.
8. Carey Foster’s bridge – Verification of laws of resistance.
9. Lee’s Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

WORKSHOP
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

COURSE OBJECTIVES:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60° pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bench vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S.K. and A.K. Choudary, Media Promoters and Publishers Pvt. Ltd

ENGLISH LANGUAGE LAB
(Common to All Branches)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1. Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012.
2. Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3. A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

NCC/NSS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE
TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of the course, students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in 'C' Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

II/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

INFORMATION TECHNOLOGY

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 2101	Data Structures	4	3	1	--	4	30	70	100
B16 EC 2103	Elements of Electronics Engineering	4	3	1	--	4	30	70	100
B16 ENG 2102	Discrete Mathematical Structures	4	3	1	--	4	30	70	100
B16 IT 2102	Object Oriented Programming Using C++	4	3	1	--	4	30	70	100
B16 IT 2103	Digital Logic Design	4	3	1	--	4	30	70	100
B16 ENG 2103	Environmental Studies	2	3	1	--	4	30	70	100
B16 IT 2104	Data Structures Lab.	2	--	--	3	3	50	50	100
B16 IT 2105	Object Oriented Programming Lab Using C++.	2	--	--	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 ENG 2105	Industry Oriented Training.	1	--	--	2	2	50	--	50
Total		29	19	7	8	34	380	570	950

DATA STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Assess how the choice of data structures and algorithm design methods impacts the performance of programs.
2. Choose the appropriate data structure and algorithm design method for a specified application.
3. Solve problems using data structures such as linear lists, stacks, queues, binary trees, heaps binary search trees, and graphs and writing programs for these solutions.

Course Outcomes:

1. Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithm.
2. Demonstrate different methods for traversing trees.
3. Compare alternative implementations of data structures with respect to performance.
4. Discuss the computational efficiency of the principal algorithms for sorting and searching

SYLLABUS

Introduction to Data Structures: Review of C Programming, Abstract Data Types, Meaning and Definition of Data Structures, Arrays in C, Recursive Definition and Processes, Recursion in C, Simulation of Recursion, Efficiency of Recursion.

Stacks: Stack as an Abstract Data Type, Primitive Operations, Implementing Stack Operations using Arrays, Infix, Postfix and Prefix: Definitions, Evaluation and Conversions.

Queues: Queue as an Abstract Data Type, Types of Queues, Operations, Implementation using Arrays.

Linked List: Operations, Implementation of Stacks, Queues and priority Queues using Linked Lists, Circular Lists- Insertion, Deletion and Concatenation Operations, Stacks and Queues as Circular Lists, Doubly Linked Lists.

Trees: Binary Trees - Definitions and Operations, Binary Tree Representation- Node Representation, Implicit array Representation, Binary Tree Traversal, Threaded Binary Trees and their Traversal, Trees and their Applications; Tree Searching: Insertion and Deletion of a node from a Binary Search Tree.

Searching: Basic Searching Techniques- Dictionary as an Abstract Data Type, Algorithmic Notation, Sequential Searching and its Efficiency, Binary Search, Interpolation Search.

Sorting: General Background- Efficiency, Asymptotic Notations, Efficiency of Sorting, Bubble Sort and Quick Sort, Efficiency of Quick Sort, Selection Sort, Binary Tree Sort, Heap Sort, Insertion Sort, Shell Sort, Address calculation Sort, Merge and Radix Sort.

Graphs and Their Application: Definition of Graphs, Representation of Graphs, Transitive closure, Warshall's Algorithm, Shortest-Path Algorithm, Linked Representation of Graphs, Topological Ordering of nodes, Graph Traversal and Spanning Forests- Undirected Graphs and their traversals, Minimal Spanning Trees.

Text Books:

1. Data Structures Using C and C++ Yadidiah Langsam, Moshe J. Augennstein and Aaron M. Tanenbaum Prentice Hall of India (2nd Edition).
2. Data Structures, Algorithms and Applications with C++, Sahani Mc-Graw Hill.

Reference Books:

1. Data Structures with C by Seymour lipschutz, Schaum Outline series, 2010.
2. Data Structures using C by R. KrishnaMoorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give the exposure to the students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED and tunnel diode.
3. To give the exposure to the students on rectifies circuits using diodes.
4. To give the exposure to the students on basics of BJT, JFET and MOSFET and biasing of BJT and FETs
5. To give the exposure to the students on the analysis of transistor at low and high frequencies.

Course Outcomes:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJTs and FETs
2. Use this knowledge to analyze and design basic electronic application circuits.
3. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport phenomenon in semiconductors:

Intrinsic and Extrinsic semiconductors, Charge densities in semiconductors, Drift and Diffusion currents, Hall effect, Mass action law.

P-N Junction Diode :

Basic operation and V-I characteristics of semiconductor diode, diode current equation, Zener diode, LED, Photo diode and tunnel diode (*Introductory treatment only*)

Diode Rectifiers:

Half wave and full wave rectifiers with and without filters, Bridge Rectifier expressions – Ripple factor, Efficiency, capacitor filters

Bipolar Junction Transistor :

Introduction, construction, basic operation, modes of operation-Active , cutoff and saturation, Transistor circuit configurations- CE, CB, CC – input and output Characteristics in various configurations (*Introductory treatment only*).

Transistor Biasing and Thermal Stabilization :

Transistor Biasing, Thermal runaway, Stabilization, Different methods of Biasing- Fixed bias, Collector feedback bias, self bias, Bias Compensation.

Transistor Amplifiers :

CE, CB, CC amplifier h-parameter model for Transistor amplifier

Field Effect Transistors :

Junction Field effect Transistors (JFET)- JFET characteristics, JFET parameters, JFET biasing, MOSFETS- Depletion and Enhancement MOSFET.

Text Books :

1. Electronic Device and Circuits by Sanjeev Guptha

Reference Books :

1. Integrated Electronics – Millman & Halkias

DISCRETE MATHEMATICAL STRUCTURES
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand mathematical arguments using logical connectives and quantifiers and verify the validity arguments using propositional, predicate logic & truth tables.
2. To know about the concepts of counting techniques
3. To know how to solve recurrence relations.
4. To understand various types of relations and discuss various properties of the relations.
5. To understand the concepts in graphs and trees.
6. To learn about Groups, Lattices and their properties, Boolean algebra and its importance in computer field and minimization of Boolean expressions.

Course Outcomes:

At the end of the course students will be able to

1. Rewrite the mathematical arguments using logical connectives and quantifiers and verify the validity of the arguments using propositional and predicate logic.
2. Solve different counting problems.
3. Solve the recurrence relations which occur in many fields.
4. Identify and give examples of various types of relations and describe various properties of relations.
5. Determine isomorphism of graphs and utilize the concepts in graphs & trees in their fields.
6. Understand the importance of Groups, lattice structures and their diagrammatic representations and also the importance of Boolean algebra in computer science.

SYLLABUS

Introduction :

Sets – Operations on sets – Logic : Logical inferences, Methods of proof of an implications– First order logic and other proof methods-Rules of inference for quantified propositions-mathematical induction.

Elementary Combinatorics & Recurrence relations :

Basics of counting – Combinations and Permutations – their enumeration with and without repetition - Principle of Inclusion and Exclusion and its applications, Generating functions of sequences - Calculating their coefficients-Recurrence relations-solving recurrence relations-method of characteristic roots-Non-homogeneous recurrence relations and their solutions.

Relations and Diagraphs :

Relations and directed graphs-Special properties of binary relations-equivalence relations-Ordering relations-operations on relations-Paths and closures-Directed graphs and Adjacency matrices.

Graphs Theory :

Basic concepts – Isomorphism – sub graphs - planar graphs - Euler's formula -Multi graphs and Euler Circuits - Hamiltonian graphs – Graph coloring and Chromatic number – Four color theorem - Trees and their properties – definitions of different tree structures.

Groups :

Definitions of Binary operation, Algebraic Structure, Semi-group, Monoid, Group and Abelian group.

Lattices :

Lattices and Properties of lattices – lattices as partially ordered sets – sublattices – Direct product and Homomorphisms - Isomorphisms – Modular lattices Distributive lattices – Complemented lattices.

Boolean Algebra :

Definition – Sub algebra – Direct product – Homomorphisms – Isomorphisms – Boolean functions – Representation of Boolean functions – Minimizations of Boolean functions using K-maps.

Text Books :

1. For the first four topics, scope and treatment as in “Discrete Mathematics for computer scientists and mathematicians” by Joe. L .Mott, Abraham Kandel & T.P. Baker, Prentice Hal of India Ltd, New Delhi.
2. For the topics five and six, scope and treatment as in “Discrete mathematical structures with applications to computer science” by J.P. Trembly & R. Manohar, Tata McGraw-Hill Publishing company, New Delhi.

Reference Books :

1. “Discrete mathematics and its applications” by Keneth. H. Rosen, Tata McGraw-Hill Publishing Company, New Delhi.
2. “Discrete Mathematics” by Richard Johnson Baug, Pearson Education, New Delhi.
3. “Discrete and Combinatorial Mathematics” by Ralph. G. Grimaldi, Pearson Education, New Delhi.

OBJECT ORIENTED PROGRAMMING USING C++

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

On completing this course student will be able to

1. Understand the syntax and principles of Object oriented programming language, and to programs using control statements, classes and interfaces.
2. Design and development of secure and extendable C++ applications.
3. Understanding the concepts of oops, different predefined classes and packages
4. Understand the concepts of polymorphism

Course Outcomes:

1. Able to outline and describe difference between OOP and POP.
2. Able to recognize and differentiate classes and objects , construct and apply Classes and Objects for real time applications.
3. Able to recognize and construct constructors and destructors and can apply usage of inheritance.
4. Able to understand polymorphism concepts and can apply real time applications.
5. Able to illustrate, identify and apply generic programming concepts with exception handling.

SYLLABUS

Basic Concepts Of OOP: Procedural Paradigms, Object Oriented Paradigm, OOP Principles and Terminology, OOP benefits, Procedure and Object Oriented programming languages, advantages and disadvantages.

Introduction to C++ : Basic Structure C++ Program , variable and Constants, Symbolic Constants , basic data types and derived data type , variable declaration , dynamic initialization, type modifiers, type casting, i/o statements in C++, operators and example programs, Control Structures- Programs using all control structures and statements, Functions: Function Prototypes, Function Components, Returning values from functions, actual and formal arguments, parameter passing methods, Inline functions,

Classes and Objects: Introduction to class, class definition, class specification, Member functions, data members, access specifiers, scope resolution operator, Object definition and creation, array of objects, pointers, Pointers to objects, this pointer, dynamic allocation operator, friend functions, const and volatile functions, static members, nested classes, local classes.

Constructors and destructors: Definition of constructor and destructor, default constructor, parameterized constructor, copy constructor, constructor with dynamic allocation, explicit constructor

Inheritance: Definition, base class, derived class, using access specifiers in inheritance, Types of Inheritance, protected data with private inheritance, constructor in derived and base class, abstract classes.

Virtual functions and Polymorphism: Function overloading, arrays and strings, Operator overloading through unary and binary operator, Friend functions, Assignment operator, Stream operator overloading and type conversion; Virtual functions, Pure Virtual function, Dynamic polymorphism, Virtual destructor, Virtual base class, Dynamic casting, Cross casting, Down casting, Program development.

Streams and Files in C++: Stream Classes, Formatted and unformatted data, manipulators, user defined manipulators, file streams, file pointer manipulation; file open and close, file handling, random access, object serialization, name spaces, std namespaces, ANSI string objects and standard template library.

Templates, Exception handling: Class templates, Function templates, Member function templates, Exception handling - try-catch-throw paradigm, exception specification, terminate and un expected functions- uncaught exception, exception handling mechanism, multiple catch, nested try, Rethrowing the exceptions

Text Books:

1. Object Oriented Programming through C++ by Robot Laphore.

Reference Books:

1. Object Oriented Programming in C++: N. Barkakati, PHI
2. Object oriented Programming using C++: E. Balagurusamy, PHI.
3. The Complete reference in C++ by Herbert Shieldt, TMH
4. The C++ Programming Language by B. Stroustrup, Pearson Education

DIGITAL LOGIC DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand Number Systems and conversions.
2. To introduce the basic principles for design of combinational circuit and sequential circuits.
3. To learn simple digital circuits in preparation for computer engineering.
4. To design and learn memory elements and system memory.

Course Outcomes:

A student who successfully fulfills the course requirements will have demonstrated:

1. An ability to define different number systems, binary addition and subtraction, 2's complement representation and operations with this representation.
2. An ability to understand the different Boolean algebra theorems and apply them for logic functions.
3. An ability to define the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
4. An ability to define the following combinational circuits: multiplexer, de-multiplexers encoders/decoders, comparators, arithmetic-logic units; and to be able to build simple circuits.
5. An ability to understand asynchronous and synchronous sequential circuits, like counters and shift registers.
6. An ability to understand memories like RAM and ROM, Programmable Logic Array and Programmable Array Logic.

SYLLABUS**Binary Systems:**

Digital Systems. Binary Numbers. Number Base Conversions. Octal and Hexadecimal Numbers. Complements. Signed Binary Numbers. Binary Codes. Binary Storage and Registers. Binary Logic

Boolean Algebra and Logic Gates:

Basic Definitions. Axiomatic Definition of Boolean Algebra. Basic Theorems and Properties of Boolean Algebra. Boolean Functions. Canonical and Standard Forms. Other Logic Operations. Digital Logic Gates. Integrated Circuits.

Combinational Logic Design, Gate-Level Minimization:

The Map Method. Four- Variable Map. Five-Variable Map. Product of Sums Simplification. Don't-Care Conditions. NAND and NOR Implementation. Other Two- Level Implementations. Exclusive-OR Function. Hardware Description Language(HDL).

Combinational Logic:

Combinational Circuits. Analysis Procedure. Design Procedure. Binary Adder-Subtractor. Decimal Adder. Binary Multiplier. Magnitude Comparator. Decoders. Encoders. Multiplexers. HDL For Combinational Circuits.

Sequential Logic Design, Synchronous Sequential Logic:

Sequential Circuits .Latches .Flip-Flops. Analysis of Clocked Sequential Circuits. HDL For Sequential Circuits. State Reduction and Assignment. Design Procedure.

Registers and Counters:

Registers. Shift Registers. Ripple Counters. Synchronous Counters. Other Counters. HDL for Registers and Counters.

Memory and Programmable Logic:

Introduction. Random-Access Memory. Memory Decoding, Error Detection and Correction. Read-Only Memory. Programmable Logic Array. Programmable Array Logic. Sequential Programmable Devices.

Text Books:

1. Digital Design, 3rd Edition, M. Morris Mano, Pearson Education.

Reference Books:

1. Digital Logic Design Principles, Norman Balabanian & Bradley Carlson, John Wiley & Sons (Asia) Pvt. Ltd., 2002.
2. Fundamentals of Digital Logic with VH.

ENVIRONMENTAL STUDIES
(Common to CIV, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs	Credits	:2

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a “CONCERN AND RESPECT FOR THE ENVIRONMENT”

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness.
Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity, Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR,10th ed, Thomson publishers, 2004

DATA STRUCTURES LAB

Lab : 3 Periods	Sessionals : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives:

1. To implement stacks and queues using arrays and linked lists.
2. To develop programs for searching and sorting algorithms.
3. To write programs using concepts of various trees.
4. To implement programs using graphs.

Course Outcomes:

1. Student will be able to write programs to implement stacks and queues.
2. Ability to implement various searching and sorting techniques.
3. Ability to implement programs using trees and graphs.

SYLLABUS

Implement the following programs using C-Language.

1. Write a program for sorting a list using Bubble sort and then apply binary search.
2. Write a program to implement the operations on stacks.
3. Write a program to implement the operations on circular queues.
4. Write a program for evaluating a given postfix expression using stack.
5. Write a program for converting a given infix expression to postfix form using stack.
6. Write a program for implementing the operations of a dequeue
7. Write a program for the representation of polynomials using circular linked list and for the addition of two such polynomials
8. Write a program for quick sort
9. Write a program for Merge sort.
10. Write a program for Heap sort
11. Write a program to create a binary search tree and for implementing the in order, preorder, post order traversal using recursion
12. Write a program for finding the transitive closure of a digraph
13. Write a program for finding the shortest path from a given source to any vertex in a digraph using Dijkstra's algorithm
14. a) Write a program for finding the Depth First Search of a graph.
b) Write a program for finding the Breadth First Search of a graph

Reference Books:

1. Data Structures using C by Tanenbaum

Code: B16 IT 2105

OBJECT ORIENTED PROGRAMMING LAB USING C++

Lab : 3 Periods	Sessionals : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives:

1. To develop programs using basic OOPS concepts such as classes and objects.
2. To implement programs using Inheritance concepts.
3. To implement programs using Exception handling.
4. To develop programs using operator overloading concepts.

Course Outcomes:

1. Student will be able to use OOPs concepts.
2. Ability to apply Inheritance concepts to several problems.
3. Ability to use Exception Handling concepts.

Implement the following Programs using C++

1. Write a Program that implements stack operations using classes and objects.
2. Write a Program performing complex number addition using friend functions.
3. Write a Program for complex number addition using operator overloading.
4. Write a Program to perform string operations by overloading operators.
5. Write a Program on hierarchical inheritance showing public, private and protected inheritances.
6. Write a Program for computation of student's result using hybrid inheritance.
7. Write a Program implementing bubble-sort using templates.
8. Write a Program on virtual functions.
9. Write a Program for handling Push On Full and Pop On Empty Exceptions for a Stack.
10. Write a Program for copying one file to another file using streams.
11. Write a Program for writing and reading a class object to a file.
12. Write program to implement
 - A. One catch block and all Exceptions
 - B. Using multiple Catch Blocks
13. Write a program to implement the finally Block.
14. Write a program to implement pointers to a derived class and virtual base classes.
15. Write a program to implement conversion of objects between different classes using conversion functions.
16. Write a program to implement function overloading- with various data types, with different number of arguments.
17. Write a program to evaluate mixed mode expressions and implicit type conversions.
18. Write a program to show that there is ambiguity in Multiple Inheritance.
19. Write a program to implement a virtual destructor.

20. Write a program to mimic a bank management system (user logins, requests for withdraw/credit, system verifies whether enough balance is available, update the account summary, etc.)

Reference Books:

1. Object Oriented Programming in C++: N. Barkakati, PHI
2. Object oriented Programming using C++: E. Balagurusamy, PHI.
3. The Complete reference in C++ by Herbert Shieldt, TMH
4. The C++ Programming Language by B. Stroustrup, Pearson Education

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	: 1 Period	Sessionals	: 50
Tutorial	: 1 Period	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to

1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvement and motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts.
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informal contexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills

PPT

Describing event/place/thing

Picture Description

Extempore

Debate

Telephonic Skills

Analyzing Proverbs

Vocabulary

Affixes

Pairs of Words

Reading Skills

Reading Comprehension

Reading/Summarizing News Paper Articles

Writing Skills

Designing Posters

Essay writing

Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated based on Presentation Skills – 30M, Project 20M)

**INDUSTRY ORIENTED TRAINING
(WEB Development)
(Common to CSE & IT)**

Lab: 2 Periods**Sessionals : 50****Exam : 3 Hrs.****Credits : 1****Course Objective:**

Web development technologies such as HTML, CSS, JavaScript and others are at the core of all modern online systems. The objective of this course is to expose & train the students on web programming concepts and develop the ability to build web sites with dynamic presentation.

Course Outcomes:

Upon completion of this web development training course, able to:

1. Design and develop basic web pages using HTML.
2. Apply cascading style sheets to web pages in order to separate form from content.
3. Understand & Apply basic control of elements with JavaScript.
4. Understand the basic concepts of PHP scripting
5. Able to design & complete a project by applying above all the concepts.

Syllabus: Industry Oriented Applications on following topics.

HTML:- HTML Introduction, HTML Basic Tags, HTML Lists, HTML Tables, HTML Images, HTML Links & Navigation, HTML Forms.

CSS:- CSS Introduction, CSS Properties - Controlling Fonts, CSS Properties - Text Formatting, Selectors - id and class, Pseudoclasses, CSS for Links, CSS for Lists, CSS for Tables.

JAVA SCRIPT:- JavaScript Introduction, Empty Field Validation Example, Name & Numbers Only Validation Example, Email Validation Example, innerHTML Error Display Example.

PHP:- Installation of Wamp Server, PHP Introduction, Creating PHP Script, Running PHP Script, PHP Numeric Variables, Sample PHP Programs

MINI PROJECT

(Note: Total Marks will be evaluated based on Continuous evaluation - 25 Marks, Mini Project- 25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

INFORMATION TECHNOLOGY

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 2201	Operating Systems	4	3	1	--	4	30	70	100
B16 IT 2202	Computer Organization	4	3	1	--	4	30	70	100
B16 IT 2203	Microprocessors	4	3	1	--	4	30	70	100
B16 IT 2204	Data Communications	4	3	1	--	4	30	70	100
B16 IT 2205	Operations Research	4	3	1	--	4	30	70	100
B16 IT 2206	Java Programming	4	3	1	--	4	30	70	100
B16 IT 2207	Java programming Lab	2	--	--	3	3	50	50	100
B16 IT 2208	Digital Electronics & Microprocessors Lab	2	--	--	3	3	50	50	100
B16 IT 2209	Python Programming	1	1	--	1	2	50	--	50
B16 ENG 2203	Industry Oriented Training	1	--	--	2	2	50	--	50
Total		30	19	6	9	34	380	520	900

Code: B16 IT 2201

OPERATING SYSTEMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand evolution of Operating System.
2. To understand operating system as a layer of abstraction above physical hardware that facilitates usage convenience and efficient resource management of computer system resources.
3. To learn design and implementation of policies and mechanisms for OS subsystem.
4. To investigate case studies to understand the design philosophies / paradigm for popular multiuser or single user operating system.

Course Outcomes:

1. The student understands OS evolution, its structure and services provided by it.
2. Learn process life cycle, process scheduling objectives, policies and mechanisms, process synchronization, inter process communication, deadlocks and other process subsystem related concepts.
3. Learn memory hierarchy, allocation and de-allocation policies and mechanism for main and auxiliary memory, file system design and implementation issues.
4. Investigate UNIX/ LINUX and Windows OS platforms w.r.t similarities and differences in design philosophies.

SYLLABUS

Introduction to Operating Systems:

Over View of Operating Systems, Types of Operating Systems, Operating System Structures, Operating System Services, System Calls, Virtual Machines, Operating System Design and Implementation.

Process Management:

Process Concepts, Operations on Processes, Cooperating Processes, Threads, Inter Process Communication, Process Scheduling, Scheduling Algorithms, Multiple - Processor Scheduling, Thread Scheduling.

Process Synchronization:

The Critical Section Problem, Peterson's Solution, Synchronization Hardware, Semaphores, Classical Problems of Synchronization, Critical Regions, Monitors.

Deadlocks:

System Model, Deadlock Characterization, Methods For Handling Deadlocks, Deadlock Prevention, Avoidance, Deadlock Detection, Recovery from Deadlocks

Memory Management:

Logical versus Physical Address, Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, Virtual Memory, Demand Paging, Page Replacement, Allocation of Frames, Thrashing, Memory-Mapped files

File Systems, Implementation, and Secondary-storage Structure:

Concept of a file, Access Methods, Directory Structure, Protection, File System Structure, Allocation Methods, Free Space Management, Directory Management, Device Drivers, overview of Mass-storage structure, Disk structure, disk attachment, disk scheduling, swap-space management.

Case study: Overview of LINUX, Windows Operating systems

Text Book:

1. Operating Systems, Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, John Wiley Publ., Seventh Edition.

Reference Books:

1. Modern Operating Systems, Andrew S. Tanenbaum, 2nd edition, 1995, PHI.
2. Operating Systems, William Stallings 5th Edition - PHI
3. Operating Systems: A Design-Oriented Approach, Charles Crowley, Tata Hill Co., 1998 edition.

COMPUTER ORGANIZATION

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study about structure and functional components of a computer.
2. Understanding the hierarchical organization of a computer system which consists of
3. instruction set of commands.
4. Learn about the architecture of a computer from a programming view.
5. To design a balance system that minimizes performance and utilization of all elements.

Course Outcomes:

1. Apply the basic knowledge about Digital logic to the Functional components of computer.
2. Students will be able to Describe the major components of a computer.
3. Students will be able to classify different Computer Instructions.
4. Students will be able to Describe Instruction set architecture.
5. Recognize the importance of peripheral devices.
6. Students should be able classify Computer memories

SYLLABUS**Register Transfer and Micro operations:**

Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design:

Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Input- Output and Interrupt, Complete Computer Description, Design of Basic Computer, Design of Accumulator Logic.

Micro programmed Control:

Control Memory, Address Sequencing, Micro program Example, Design of Control Unit.

Central Processing Unit:

Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer(RISC) 23

Pipeline and Vector Processing:

Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISK Pipeline, Vector Processing, Array Processors.

Input/output Organization:

Peripheral Devices, I/O interface, Asynchronous data transfer, Modes of transfer, priority Interrupt, Direct memory access, Input-Output Processor (IOP), Serial Communication.

Memory Organization:

Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory, Memory Management Hardware.

Text Book :

1. Computer System Architecture, M. Morris Mano, Prentice Hall of India Pvt. Ltd., Third Edition, Sept.2008.

Reference Books :

1. Computer Architecture and Organization, William Stallings, PHI Pvt. Ltd., Eastern Economy Edition, Sixth Edition, 2003.
(Computer Organization and Architecture-Designing of Performance)
2. Computer Organization and Architecture, Linda Null, Julia Lobur, Narosa Publications, Third Edition ,2003.
3. Computer Architecture and Organization, John. P. Hayes, Third Edition, Tata Mc Grawhill International Edition,1998.

MICROPROCESSORS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To discuss the architectures of 8085, 8086 microprocessors, their instruction sets and related ALP programs.
2. To discuss interfacing semiconductor memories, interfacing peripheral to Intel 8086.
3. To study interfacing data converters to 8086
4. To study different programming techniques to implement on Microsoft Assembler.

Course Outcomes:

1. Understand the basic architectures of 8085 and 8086 microprocessors.
2. Ability to write ALP using instruction sets of 8085.
3. Understand the various interfacing concepts.
4. Ability to write ALP using instruction sets of 8086.
5. Understand how to interface peripherals with 8086.

SYLLABUS

Internal Architecture functional/signal description of 8085 microprocessor, Instruction set, Addressing modes and programming in 8085.

Timing diagram, counters and delays, stacks and subroutines and Interrupts in 8085

Classification and interfacing semiconductor memories with 8085 MPU. Interfacing characteristics of IO devices, IO device addressing methods.

Interfacing peripherals to INTEL 8085: Parallel IO interface-8255, Serial IO Interface-8251, Timer Interface-8253.

Interfacing peripherals to INTEL 8085: Keyboard/Display Interface-8279, Interrupt controller Interface-8259.

The 8086 Microprocessor architecture , Internal Architecture & functional /signal description of 8086, segmented memory, Maximum 7 Minimum mode of 8086.

Introduction set and programming the 8086: Addressing modes, Instruction set and assembly language programming techniques with 8086.

Text Books

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The Mcgraw-Hill companies.

Reference Books

1. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
2. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfacing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
3. Microprocessors and Interfacing. Programming and hardware, 2ne Edition, Douglass V. Hall. MH Edition , 1999.

DATA COMMUNICATIONS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study basics of data communication systems.
2. To study the various types of transmission media.
3. To study the various hardware concepts related to data communications.
4. To discuss about modem and multiplexing techniques.

Course Outcomes:

1. Students will have the ability to use Data Communications and Networking Protocols and protocol architectures
2. Students will have the ability to develop communication models for providing data transmission facility
3. Students will have the ability to outline Data Communication terminology
4. Students will have the ability to classify various transmission media
5. Students will have the ability to discriminate various types of signals for data transmission and ability to describe data encoding techniques
6. Students will have the ability to describe data communications interface
7. Students will have the ability to apply various flow control , error control techniques of data link control protocols
8. Students will have the ability to use various data communication terminals and processing hardware
9. Students will have the ability to demonstrate multiplexing techniques

SYLLABUS**Introduction to Data Communications:**

A Communications Model, Data Communications and Data Communications Networking, Protocols and Protocol Architecture, Characteristics of Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments.

Transmission Media:

Guided Transmission Media, Wireless Transmission. **Data Encoding:** Digital Data-Digital Signals, Digital Data-Analog Signals, Analog Data-Digital Signals, Analog Data-Analog Signals.

Data Communication Interface:

Asynchronous and Synchronous Transmission, Line Configurations, Interfacing. Data Link Control Flow Control, Error Detection, Error Control, High-Level Data Link Control (HDLC).

Data Communications Hardware:

Terminals: Introduction, Basic Terminal Components, Enhanced Terminal Components, General-Purpose Terminals, Remote Job Entry Terminals, Transaction Terminals, Clustering of Terminal Devices.

Communication Processing Hardware:

Introduction, Switching Processors, Multidrop Lines, Multiplexers, Concentrators, Front-End Processors

Multiplexing:

Frequency-Division Multiplexing, Synchronous Time-Division Multiplexing: Characteristics, TDM Link Control, Digital Carrier Systems, Statistical Time-Division Multiplexing: Characteristics.

Text Books:

1. William Stallings, Data and Computer Communications, 7th Edition, PH/Pearson Edu. Inc.,
2. Mary E.S. Loomis, Data Communications, PHI-N.J., 1983 (Chapter 3, Chapter 5)

Reference Books:

1. Behrouz A. Forouzan, Data Communications and Networking, 3rd Edition TMH, 2004
2. William A. Shay, Understanding Data Communications & Networks, 2nd Edition Thomson-Brooks/Cole –Vikas Publishing House, 1999.
3. Michale A. Miller, Data & Network Communications, Thomson/Delmar –Vikas Pub. House, 2000

OPERATIONS RESEARCH

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To discuss about basic Operation Research concepts , Formulation of LPP and its solution using graphical method.
2. To discuss about standard form of LPP. solving LPP using various methods.
3. To study the various solutions of transportation problems and assignment problems.
4. To discuss about PERT and CPM charts
5. To discuss about replacement problems, inventory problems and game theory.

Course Outcomes:

1. Ability to solve LPP problems using various methods.
2. Ability to solve transportation and assignment problems using several methods.
3. Analyze the PERT and CPM charts
4. Ability to solve replacement problems and game theory problems.

Overview of Operations Research, Types of OR Models , Phases of Operations Research–OR Techniques, Introduction to Linear Programming, Formulation of Linear Programming Problem, Graphical Solution; Graphical Sensitivity Analysis,

Standard Form of LPP, Basic Feasible Solutions, Unrestricted Variables, Simplex Algorithm , Artificial Variables, Big M Method, Two Phase Simplex Method, Degeneracy, Alternative Optimal, Unbounded Solutions, Infeasible Solutions, Primal And Dual Problems And Their Relations, Dual Simplex Method

Transportation Problem as LPP, Initial Solutions, North West Corner Rule, Lowest Cost Method, Vogels Approximation Method, Optimum Solutions of TPP, Degeneracy in Transportation, Transportation Algorithms,

Assignment Problem, Assignment Problem as LPP, Hungarian Method, Travelling Salesman Problem, Solutions of TSP, Sequencing Problems, N-Jobs Two Machine Problems, N-Jobs K Machines Problems, Two-Jobs M- Machine Problems, Crew Scheduling Problems

Network Representation of A Project, CPM and PERT, Critical Path Calculations, Time – Cost Optimizations, PERT Analysis and Probability Considerations, Resource Analysis in Network Scheduling.

Inventory Control- Inventory-Factors Effecting Inventory-EOQ, Inventory Problems With and Without Shortages, Inventory Problems with Price Breakups, Multi Item Deterministic Problems. Probabilistic Inventory Problems

Game Theory: Two Person Zero Sum Games, Mixed Strategy Games and Their Algorithms.

Text Books:

1. Operations Research, Kanti Swaroop, P.K. Gupta, Man Mohan, Sulthan Chand& Sons Education
2. Publishers Operations Research – An Introduction, Handy A Taha – Pearson Education .

Reference Books:

1. Operations Research Panneer Selvan Prentice Hall Of India.
2. Operations Research By S.D Sharma
3. Introduction To Operations Research, F.S. Hiller, G.J. Liberman, TMH
4. Operations Research, Richard Bronson, Schaum's Series

JAVA PROGRAMMING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To know basic principles of Object Oriented Programming in the context of java programming language.
2. To study different types of arrays to design programs.
3. To study and apply different types of java concepts like multithreading ,packages ,exception handling ,interfaces and design programs using these concepts.
4. To study and apply various features of AWT components.
5. To know the basic concepts of networking in the context of java programming.

Course Outcomes:

1. Ability to define different procedural and object oriented concepts and will be able to apply and differentiate between them.
2. Ability to define, understand and differentiate different types of arrays and apply them.
3. Ability to recognize various concepts of java and develops the programs using them.
4. Ability to identify and differentiate the various features of AWT components to construct container based programs.
5. Ability to describe and explain the concept of networking.

SYLLABUS

Fundamentals: HTML, OOP Concepts, Comparing JAVA with C & C++,JAVA Programming language Syntax, Variables, Data types, statements and expressions.

Control Statements: If else, for, while, and do while loops, Switch statements.

Arrays & Structures: One Dimensional & Two Dimensional Arrays, Named Structures.

Functions: Parameter Passing, Static Modifier.

Features of JAVA: Classes and Interfaces, Threads and multithreaded programming, Exception handling, Introduction to packages, Math package, Lang package, Util package.

Applet Programming: Events, Event driven programming, Events like buttons, mouse, keyboards etc., Applets, Applets package, Fonts, colours, Graphics, images. AWT components, layout managers, writing event driven program using components.

Networking:

Networking Basics: Socket overview, Client/Server, Reserved sockets. Proxy servers, Internet addressing; Java and the net, Inet address, TCP/IP client sockets, URL, URL connection, TCP/IP server sockets, Datagrams.

Textbooks:

1. Introduction to Java programming, a primer II, Balaguruswamy.
2. Java Complete Reference , Herbt Schild.

Reference Book:

1. Introduction to Java programming, Daneal/Young PHI

JAVA PROGRAMMING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To demonstrate the compilation and interpretation of java programs
2. To make practice on different Object Oriented Programming Concepts
3. To write java programs to implement java concepts
4. To write java programs to implement applet programming
5. To write java programs to generate and handle event handling programs

Course Outcomes:

1. Students will be able to understand compiling and interpreting programs.
2. Students will be able to Explore features of Object Oriented Programming.
3. Students will be able to implement various java concepts
4. Students will be able to Develop java Programs to implement applets
5. Students will be able to Develop java Programs to generate and handle events.

LIST OF PROGRAMS

1. (a) Program to display the area of a rectangle.
 (b) Program to find Sum of series $1+x+x^2+x^3+\dots$
2. (a) Write a class to display the area of rectangle and inherit this class into other class which is displaying perimeter of a rectangle and implement.
 (b) Write a class to add three no's inherit this class into other class to add five no's and implement it.
3. (a) Write a program to print the path, filename and extension for a given path of a file.
 (b) Write a program to receive two command line arguments check whether they are equal or not.
4. (a) A program to take two arguments and divide the first argument with second argument and display the result. Displays the error message if divide by zero without abnormal exit.
 (b) A program to accept more than one string and arrange them in alphabetical order.
 (c) Write a program to display simultaneously output of even and odd numbers starting from one to specified number.
5. Write a program to accept data from keyboard and write it into a file.
6. Write a java program to implement stack & Queue operations.
7. Write a program to draw line and circle using mouse.
8. Write an applet program for drawing the bar chart.
9. Write an applet program to design a calculator for implementing basic functions like +, -, *, /.
10. Write a program to check active ports in system.

Reference Books:

1. Introduction to Java programming, a primer II, Balaguruswamy.
2. Java Complete Reference , Herbt Schild.
3. Introduction to Java programming, Daneal/Young PHI

DIGITAL ELECTRONICS AND MICROPROCESSORS LAB

Lab : 3 Periods

Exam : 3 Hrs.

Sessionals : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To learn about logic gates, half adders, full adders and flip-flops.
2. To learn about the microprocessor programming.
3. To learn about the microprocessor interfacing with stepper motor, elevator.
4. To learn about the Microprocessor interfacing with R-2R ladder network.

Course Outcomes:

1. The student understands the logic gates, half adders, full adders and flip-flops to design a circuit.
2. The student develops the skill of writing microprocessor programming with 8085.
3. The student understands the interfacing of microprocessor with stepper motor, R-2R ladder.
4. The student will be able to write ACP for 8086.

LIST OF PROGRAMS

DIGITAL EXPERIMENT

Verification of Truth tables of OR, AND, NOT, NAND, NOR, EX-OR gates (by using 7400-series)

Construction of gates using NAND, NOR gates.

Construction of Half and Full adders and verifying their truth tables.

Operation and verifying truth tables of flip-flops-RS, D and JK using IC's

Up/Down counters using JK flip-flops.

4-bit shift right and left registers using JK flip-flops.

MICROPROCESSORS: 8085

Binary Addition of 'N' 8-bit numbers.

Binary to BCD conversion

Arranging –Ascending/descending order

To find the largest /smallest numbers in the array.

ASCII to HEXA & HEXA to ASCII conversion.

MICROPROCESSORS: 8086

Linear Search

Factorial of a given number

To copy string from S1 to S2

To find GCD and LCD

MICROPROCESSOR INTERFACING WITH 8085

Elevator

Traffic Light.

Analog to Digital & Digital to Analog Convertors

Interrupt controller

Stepper Motor controller.

Reference Books

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The McGraw-Hill companies.
3. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
4. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfaing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
5. Microprocessors and Interfacing. Programming and hardware, 2ne Edition, Douglass V. Hall. MH Edition , 1999.

PYTHON PROGRAMMING**Theory: 1 Period.****Lab : 1 Period.****Exam : 3 Hrs.****Sessionals : 50****Credits : 1****Course Objectives:**

1. To Learn basic concepts of Python Programming Language
2. To Learn various Object Oriented Programming Concepts
3. To Learn various advanced Data structures

Course Outcomes:

At the end of the course the student can able to

1. Write programs using python programming
2. Write algorithms
3. Implement various data Structures
4. To apply object oriented mechanisms
5. To Implement various Advance data Structures like AVL trees, B-Trees, Splay trees etc

SYLLABUS

Overview, Environment Set Up, Basic Syntax, Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation, Comments, Multiple Statements on a Single Line Variable Types, Standard Data Types, Numbers (math, random, fraction) , Strings, Lists, Tuples , Dictionaries

Operators, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Decision Making :if, if-else, nested if , Loops: for, while, nested loops

Functions, Function Arguments: Required arguments, Keyword arguments, Default arguments, Variable-length arguments, The Anonymous Functions: lambda, Scope of Variables, Modules, sys, os , Date & Time

Files & its operations, Exceptions, Standard Exceptions, Assertions, The try-finally Clause, Raising an Exception, User-Defined Exceptions, Classes and objects , OOPS, Data member , Function overloading, Instance variable, Inheritance, Instance, Instantiation, Operator overloading.

HTML,CSS Basics, Data Base(SQLite), Database Connection, CRUD Application , CGI Architecture, Web Server Support and Configuration, GET and POST Methods, CGI Scripts

Project Work

Text Books:

1. Python Cookbook, D. Beazkey & B.K. Jones, O'Reilly Series, 3rd Edition,
2. Python Programming for beginner “ The comprehensive guide to Python Programming ”, 1st Edition, Adam Stark

Reference Books :

1. Kent D. Lee, Steve Hubbard, "Data Structures and Algorithms with Python", Springer Publications. Prabhanjan Narayanachar Tattar, Suresh Ramaiah, B.G. Manjunath, "A course in statistics with R", WILEY Publications.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest- 25 Marks)

INDUSTRY ORIENTED TRAINING
(Common to CSE & IT)

Lab : 2 Periods
Exam : 3 Hrs.

Sessionals : 50
Credits : 1

Course Objectives:

Students learn about

1. Linked lists, file operations and their applications.
2. Searching and Sorting algorithms.
3. Stack & Queue operations and their applications.
4. The concepts and applications of Trees and Graphs.

Course Outcomes:

Students will be able to

1. Implement the linked lists in real time applications.
2. Apply the file handling operations.
3. Apply the Searching & Sorting algorithms.
4. Implement Stack & Queue operations.
5. Implement the concepts and applications of Trees and Graphs.

Syllabus: Industry Oriented Applications on following topics.

Linear Linked Data: Singly linked list, operations on a linked list, circular linked list, double linked list, operations on double linked list

Standard Storage: Introduction to files, file types, file modes, file functions

Searching & Sorting: Linear search and Binary search, Bubble sort, Selection sort, Insertion sort, Quick sort, Heap sort, Merge sort: Worst and Average case analysis. Decision Tree Model and (worst case) Lower Bound on Sorting. Sorting in linear time- shell sort, radix sort, bucket sort, counting sort.

Stack & Queue: Stack structure, operations. Stack using linear list data. Stack using linear linked data. Queue structure, operations. Queue using linear list data. Queue using linear linked list. Circular queues.

Non Linear Data: Tree Structure and terminology, Binary Trees, Binary Tree traversals, Applications of Binary Tree, Binary Tree Operations. Priority queues, union-find sets, (augmented) interval trees, (augmented) balanced BSTs and binary indexed trees, Binary Indexed Tree or Fenwick tree, Segment Tree (RMQ, Range Sum and Lazy Propagation), K-D tree, Union Find Disjoint Set, Tries, Interval Tree

Graphs: Graphs and their basic properties- degree, path, cycle, subgraphs, isomorphism, Eulerian and Hamilton walks, graph coloring, planar graphs, trees. Breadth first search and connected components. Depth first search in directed and undirected graphs.

More Trees: Binary search trees, Operations on BST, balanced binary search trees, AVL trees, Red-Black trees, skip lists, hashing. Priority queues, heaps, Fibonacci heap, union-find, splay trees Interval trees, tries.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest- 25 Marks)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ESTD: 1980

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

III/IV B.TECH

(With effect from 2016-2017 Admitted Batch onwards)

Under Choice Based Credit System

INFORMATION TECHNOLOGY

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 3101	Computer Networks	4	3	1	-	4	30	70	100
B16 IT 3102	Web Technologies	4	3	1	-	4	30	70	100
B16 IT 3103	Formal Languages And Automata Theory	4	3	1	-	4	30	70	100
B16 IT 3104	Database Management Systems	4	3	1	-	4	30	70	100
# ELE-I	ELECTIVE-I	4	3	1	-	4	30	70	100
B16 IT 3109	Database Management Systems Lab	2	-	-	3	3	50	50	100
B16 IT 3110	Web Technologies Lab	2	-	-	3	3	50	50	100
B16 ENG 3102	Verbal & Quantative Aptitude-I	2	5	--	--	5	100	--	100
# M-I	MOOCS-I	2	4	--	-	4	100	--	100
# M-II	MOOCS-II	2	4	--	--	4	100	--	100
Total		30	28	5	6	39	550	450	1000

# ELE-I	B16 IT 3105	Principles of Programming Languages
	B16 IT 3106	Advanced Computer Architecture
	B16 IT 3107	File Structures
	B16 IT 3108	Bio Informatics
# M-I	B16 ENG 3104	Advanced Coding
	B16 IT 3111A	IOs Application Development
	B16 IT 3111B	Ruby
	B16 IT 3111C	Programming, data structures and algorithms using Python
# M-II	B16 IT 3112A	Android App Development
	B16 IT 3112B	Black chain Architecture Design and Usecase
	B16 IT 3112C	Testing Tools
	B16 IT 3112D	Machine Learning

COMPUTER NETWORKS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To make the students understanding of basic requirements of network hardware, software and its architecture.
2. Familiarize the students with layered architecture of the network software and hierarchical nature of the network physical infrastructure.
3. Study of various network interconnecting devices and other associated network hardware.
4. Introduction of advanced networking concepts and wireless and wireless sensor networks.

Course Outcomes:

1. The student must be able to understand the design and estimate the requirements for practical setup of a given network scenario and size.
2. Realize the Operation, maintenance and management of the Internet by mapping the theoretical networking concepts to the real-time network scenarios.
3. Demonstrate the applications of wireless Networks and over view of advanced networking concepts.
4. Identify different networking devices and their usage and functionality

SYLLABUS

Introduction to Computer Networks: Introduction, Network Hardware, Network Software, Reference Models, Network Examples, Internet Based Applications.

The Medium Access Control: The Channel Allocation Problem, CSMA Protocols, Collision Free Protocols, The Ethernet, Wireless LANs, Bluetooth

Network Layer: Network Layer Design Issues, Routing Algorithms, Congestion Control Algorithms, Network Layer in the Internet, IP Protocol, IP Address, Subnets, and Internetworking.

Transport layer: Transport Service, Elements of Transport Protocols, TCP and UDP Protocols, Quality of Service Model,

Application Layer: Over View of DNS, Electronic Mail, FTP, TFTP, BOOTP, HTTP Protocols

Network Devices: Over View of Repeaters, Bridges, Routers, Gateways, Multiprotocol Routers, Hubs, Switches, Modems, Channel Service Unit CSU, Data Service Units DSU, NIC, Wireless Access Points, Transceivers, Firewalls, Proxies.

Text Book:

1. Data and Computer Communications, William Stallings 7th Edition, Pearson Education, 2004

Reference Books:

1. Data Communications and Networking, Behrouz A. Forouzan, 3rd Edition, TMH, 2004
2. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose and Keith
3. W. Ross , 2nd Edition, Pearson Education, 2002
4. Computer Networks, Andrew S. Tanenbaum, 4th Edition, Pearson Education, 2003
5. An Engineering Approach to Computer Networking, S. Keshav, Pearson Education, 1997
6. Computer Networks and Internets with Internet Applications, Ddouglas e. Comer, 4th Edition, Pearson Education, 2003

WEB TECHNOLOGIES

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

On completing this course student will be able to

1. Understand the principles of Web based application development.
2. Design dynamic content in Web Pages using JavaScript.
3. Understanding the concepts of java Servlets, design applications using them.
4. Understand the concepts of Component development and design applications by establishing connections to Databases

Course Outcomes:

1. Students will be able to construct web based applications.
2. Students will be able to connect PHP to different databases.
3. Students will be able to develop CRUD based PHP application

SYLLABUS:

Introduction to HTML, Core Elements, Links and Addressing, Images, Text, Colors and Background, Lists, Tables Frames, Forms, Cascading StyleSheets.

Introduction to Java Scripts, Elements of Objects in Java Script, Dynamic HTML with Java Script

Document type definition, XML Syntax, XML Schemas, Document Object model Using XML Processors

Introduction to Servlet, Servlet Life Cycles, Servlet Basics, Tomcat Web Server, Configuring Apache Tomcat, Handling Client Request and Response, HandlingCookies, Session Tracking.

Introduction to PHP, Language Basics, Functions, Strings, Arrays.

Web Techniques, Data bases, Graphics, PDF, Dates and Times.

MYSQL Installation, Accessing MySQL Using PHP, Form Handling, Cookies, Sessions, and Authentication, Tables, Inserting Data into Tables , Selecting Data froma Table, Updating Table , Deleting data from Table, Webpage creation.

Text Books:

1. Web Programming, building internet applications, 2nd Ed., Chris Bates, WileyDreamtech.
2. The complete Reference HTML and DHTML, Thomas A. Powey. Learning Php, Mysql, Robin Nixon
3. Programming Php, Kevin Tatroe, PeterMacIntyre&RasmusLerdorf foreword by Michael Bourque.

Reference Books:

1. Internet , World Wide Web , How to program, Dietel , Nieto, PHI/PEA.
2. Web Tehnologies, Godbole, kahate, 2nd Ed., TMH

FORMAL LANGUAGES & AUTOMATA THEORY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the concepts in automata theory and theory of computation to design grammars and recognizers for different formal languages.
2. To Employ finite state machines to solve problems in computing.
3. To introduce finite state machines, context free grammars and Turing Machines and their properties as the basis for the formal expressivity of computer languages for solving linguistic decision problems.
4. To understand the concepts of decidability.

Course Outcomes:

1. Ability to think analytically and intuitively for problem solving situations in related areas of theory in computer science.
2. Ability to describe the language accepted by an automaton or generated by a regular expression or a context-free grammar.
3. Ability to Understand the functioning of Finite-State Machines, Deterministic Finite-State Automata, Nondeterministic Finite-State Automata and Pushdown Automata and Turing Machines.

SYLLABUS

Definitions of alphabet, strings, language, grammar, types of grammar, types of machines, generation of languages from grammar, construction of grammar from the given description of languages

Definition of finite state machine, Deterministic Finite Automata and Non-Deterministic Finite Automata, representations in mathematical diagram, tabular etc., id of finite state machine's, design of finite state machine from the given description, elimination of ϵ -transitions, indefinite state machine to definite state machine, optimization of finite state machine.

Conversion of regular grammar to finite state machine, finite state machine to regular grammar, discussion of pumping lemma, systematic way of construction of finite state machine.

Definition of regular expression, regular algebra, minimization of regular expressions, closure properties, construction of regular expression from the given description, regular expression to finite state machine, finite state machine to regular expression, construction of regular expression for the given finite state machine- a systematic way using Arden's theorem.

Definition of push down machine, push down machine, types of push down machine's, push down machine to context free grammar, context free grammar to push down machine, design methodology of various push down machine's, push down machine by empty stack, push down machine by final states, conversion from one type to other type, applications of push down machine's.

Parsing tree, bottom-up parsing, top-down parsing, types of context free grammar's, leftmost and right most derivations, productions, reductions, optimization of context free grammar's, elimination of ϵ -productions, unit productions, normal forms- cnf, gnf.

Definition of Turing machine, ways of representing Turing machine's- tabular form, diagram, mathematical form, quintuples etc., design of Turing machine, id of Turing machine, types of Turing machine, halting problem, church's thesis, universal Turing machine, Gödel number, definitions of recursive functions- prf, rf, decidability.

NOTE : Theorem proofs are eliminated

Text books :

1. Introduction to automata theory, languages and computation, John.E.H.P croft/ Rajeev Motwani & JD Ullman—pearson education- III edition

Reference Books :

1. Theory of computation, K.L.P.Mishra and N.Chandrasekhar, PHI .
2. Theory of computation, formal languages and automata theory, G P Saradhi Varma, B.Thirupathi Rao –Sci Tech publications.

DATABASE MANAGEMENT SYSTEMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To learn the evolution of DBMS Versus File systems, data models, and layers of abstraction.
2. To understand conceptual and physical aspects of database design.
3. To learn formal and commercial query language specifications.
4. To understand concurrency control, recovery management, and other related issues.

Course Outcomes:

1. The student will understand ER-modeling for conceptual database design and relational model.
2. The student is introduced to formal and commercial query languages: Relational Algebra, calculus and SQL.
3. The student will learn schema refinement and normalization.
4. The Student understands locking protocols concurrency control, and crash recovery methods.

SYLLABUS

Introduction: File system versus a DBMS, Advantages of a DBMS, Describing and Storing Data in a DBMS, The Relational model, Levels of abstraction, Data Independence, Transaction management, Structure of a DBMS.

Introduction to Database Design and The Relational Model: Database Design and ER Diagrams, Entities, Attributes and Entity Sets, Relationships & Relationship Sets, Additional Features of the ER Model, Conceptual Design with ER Model, Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity Constraints, Querying Relational Data, Logical Database Design: ER to Relational, Introduction to Views, Destroying/ Altering Tables and Views.

Relational Algebra and SQL: Preliminaries, Relational Algebra, The form of a Basic SQL Query, UNION, INTERSECT and EXCEPT, Nested Queries, Aggregate Operators, Null Values, Complex Integrity Constraints in SQL, Triggers and Active Databases, Embedded SQL, Dynamic SQL, JDBC.

Database Design: Schema Refinement and Normal Forms, Introduction to Schema Refinement, Functional Dependencies, Reasoning about FD's, Normal Forms, Properties of Decomposition, Normalization, Other kinds of Dependencies.

Transaction Management: The ACID Properties, Transactions & Schedules, Concurrent Execution of Transactions, Lock-Based Concurrency Control.

Concurrency Control: 2PL, Serializability and Recoverability, Introduction to Lock Management, Lock Conversions, Dealing with Deadlocks, Specialized Locking Techniques, Concurrency Control without Locking.

Crash Recovery: Introduction to ARIES, The Log, Other Recovery-Related Structures, The Write-Ahead Log Protocol, Check pointing, Recovering from a System Crash, Media Recovery.

Text Book:

1. Database Management Systems; Raghu Ramakrishnan, Johannes Gehrke 4th Edition, McGraw- Hill

Reference Book:

1. Database System Concepts; A. Silberschatz, H. Korth 5th Edition, McGraw-Hill

PRINCIPLES OF PROGRAMMING LANGUAGES (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives:

1. To learn the underlying principles and concepts of programming language.
2. To understand programming language translation process.
3. To expose students to the important paradigms of programming.
4. To understand the concepts of distributed processing and network programming.

Course outcomes:

1. Ability to compare different programming languages.
2. Ability to discuss the significant achievements in programming language history.
3. Ability to assess the programming languages in scientific manner.

SYLLABUS

Language Design Issues: Study Programming Languages, History of Programming Languages, Role of Programming Languages, Programming Environments.

Impact of Machine Architectures: Operation of a Computer, Virtual Computers and Binding Times; Language Translation Issues: Programming Language Syntax, Stages in Translation, Formal Translation Models, Recursive Descent Parsing; Modeling Language Properties: Formal Properties of Languages, Language Semantics.

Elementary Data Types: Properties of Types and Objects, Scalar Data Types, Composite Data Types Encapsulation: Structured Data Types, Abstract Data Types, Encapsulation by Subprograms, Type Definitions .Inheritance: Abstract Data Types Revisited ,Inheritance, Polymorphism

Sequence Control: Implement and Explicit Sequence Control, Sequence with Arithmetic Expressions, Sequence Control Between Statements, Sequencing with Non-arithmetic Expressions.

Subprogram Control: Subprogram Sequence Control, Attributes of Data Control, Parameter Transmission, Explicit Common Environment.

Storage Management: Elements Requiring Storage, Programmer- and System - Controlled Storage, Static Storage Management, Heap Storage Management.

Distributed Processing: Variations on Subprogram Control, Parallel Programming, Hardware Developments, Software Architecture. Network Programming: Desktop Publishing ,The World Wide Web

Text Book:

1. Programming languages – Design and Implementation by Terrence W. Pratt Marvin V. Zelkowitz.3 rd Edition, Prentice Hall of India.

Reference Books:

1. Concepts of Programming Languages by Robert L. Sebesta, 4th Edition ,Pearson Education.
2. Fundamentals of Programming Languages, Design & Implementation by SeyedH .Roosta. Vikas publications.
3. Programming Languages by Paradigm and Practice – Doris Appleby Julius J. Vendekopple Tata McGraw HillEdition.

**ADVANCED COMPUTER ARCHITECTURE
(Elective-I)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Understand the Concept of Parallel Processing and its applications.
2. Implement the Hardware for Arithmetic Operations.
3. Analyze the performance of different scalar Computers.
4. Develop the Pipelining Concept for a given set of Instructions.
5. Distinguish the performance of pipelining and non pipelining environment in a processor.

Course Outcomes:

1. Understand the Concept of Parallel Processing and its applications.
2. Implement the Hardware for Arithmetic Operations.
3. Analyze the performance of different scalar Computers .
4. Develop the Pipelining Concept for a given set of Instructions.
5. Distinguish the performance of pipelining and non pipelining environment in a processor

SYLLABUS

Fundamentals of Computer design- Technology trends- cost- measuring and reporting performance quantitative principles of computer design.

Instruction set principles and examples- classifying instruction set- memory addressing- type and size of operands- addressing modes for signal processing-operations in the instruction set- instructions for control flow- encoding an instruction set.-the role of compiler.

Instruction level parallelism (ILP)- over coming data hazards- reducing branch costs –high performance instruction delivery- hardware based speculation- limitation of ILP

ILP software approach- compiler techniques- static branch protection – VLIW approach – H.W support for more ILP at compile time- H.W verses S.W Solutions

Memory hierarchy design- cache performance- reducing cache misses penalty and miss rate – virtual memory- protection and examples of VM.

Multiprocessors and thread level parallelism- symmetric shared memory architectures- distributed shared memory- Synchronization- multi threading.

Storage systems- Types – Buses – RAID- errors and failures- bench marking a storage device- designing a I/O system.

Inter connection networks and clusters- interconnection network media – practical issues in interconnecting networks- examples – clusters- designing a cluster.

Text Book

1. Computer Architecture A quantitative approach 3rd edition John L. Hennessy & David A. Patterson Morgan Kaufmann (An Imprint of Elsevier)

Reference Books :

1. “Computer Architecture and parallel Processing” Kai Hwang and A.Briggs International Edition McGraw-Hill.
2. Advanced Computer Architectures, Dezso Sima, Terence Fountain, Peter Kacsuk, Pearson.
3. Parallel Computer Architecture, A Hardware / Software Approach, David E. Culler, Jaswinder Pal singh with Anoop Gupta, Elsevier

FILE STRUCTURES (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Provide an introduction to the fundamental file operations and storage systems.
2. Introducing fundamental concepts of file structure.
3. Introducing the most important high-level file structures tools which include indexing, co sequential processing, B trees, Hashing.
4. Applying the techniques in the design of C++ programs for solving various file management problems.

Course Outcomes:

1. Understand the fundamental concepts of file processing operations and storage structures
2. Apply object orientation concepts to manipulate records.
3. Apply concepts of sorting and merging on multiple files.
4. Analyze the sequential and indexing file accessing techniques with appropriate data structures.
5. Illustrate the usage of hashing techniques to organize file structures

SYLLABUS

File Processing Operations and Secondary Storage

Physical and logical files, opening, reading & writing and closing files in C, seeking and special characters in files, physical devices and logical files, file-related header files in C

Disks – organization, tracks, sectors, blocks, capacity, non-data overhead, cost of a disk access, Magnetic Tape – types, performance, organization estimation of tape length and data transmission times, disk vs tape, CD-ROM – CD-ROM as a file structure, physical organization, strengths and weakness of cd-roms, storage hierarchy

Byte Journey and buffer Management and File Structure Concepts

File manager, I/O buffer, I/O processing, buffer strategies and bottlenecks

A stream file, field structures, reading a stream of fields, record structures and that uses a length indicator, Mixing numbers and characters – use of a hex dump, reading the variable length records from the files

Managing records in C files and Organizing files for performance

Retrieving records by keys, sequential search, direct access, choosing a record structure and record length, header records, file access and file organization

Data compression, reclaiming space – record deletion and storage compaction, deleting fixed-length records for reclaiming space dynamically, deleting variable-length records, space fragmentation, replacement strategies.

Indexing and Indexed sequential file access and prefix B+ Trees

Index, A simple index with an entry sequenced file, basic operations on an indexed, entry sequenced file, indexes that are too large to hold in memory, indexing to provide access by multiple keys, retrieval using combination of secondary keys, improving the secondary index structure – inverted lists

Indexed sequential access, maintaining a sequence set, adding a simple index to the sequence set, the B tree, simple prefix B+ content of the index: separators instead of keys, the simple prefix tree maintenance, index set block size, internal set block size, internal structure of index set blocks, loading a simple prefix.

Hashing and Extendable hashing

Collisions in hashing, a simple hashing algorithms, hashing functions and record distributions, memory requirements, collision resolution by progressive overflow, buckets, deletions

Working of extendable hashing, implementation, deletion, extendable hashing performance

Text Book:

1. File Structures – An Object Oriented Approach with C⁺⁺ by Michael J. Folk, Bill Zoellick and Greg Riccardi,, Pearson

Reference Books:

1. File Organization & Processing by Alan I.Tharp Malloy Lithographing Inc.
2. File Systems by Thomas Harbron

BIO INFORMATICS (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. The basic objective is to give students an introduction to the basic practical techniques of bioinformatics.
2. Emphasis will be given to the application of bioinformatics and biological databases to problem solving in real research problems.
3. The students will become familiar with the use of a wide variety of internet applications, biological database and will be able to apply these methods to research problems.

Course Outcomes:

1. The students will be able to describe the contents and properties of the most important bioinformatics databases, perform text- and sequence-based
2. searches, and analyze and discuss the results in light of molecular biological knowledge.
3. The students will be able to explain the major steps in pair wise and multiple sequence alignment, explain the principle for, and execute pair wise
4. sequence alignment by dynamic programming The students will be able to predict the secondary and tertiary structures of protein sequences.

SYLLABUS

Introduction: Definitions, Sequencing, Biological sequence/structure, Genome Projects, Pattern recognition and prediction, Folding problem, Sequence Analysis, Homology and Analogy.

Protein Information Resources : Biological databases, Primary sequence databases, Protein Sequence databases, Secondary databases, Protein pattern databases, and Structure classification databases.

Genome Information Resources : DNA sequence databases, specialized genomic resources

DNA Sequence analysis : Importance of DNA analysis, Gene structure and DNA sequences, Features of DNA sequence analysis, EST (Expressed Sequence Tag) searches, Gene hunting, Profile of a cell, EST analysis, Effects of EST data on DNA databases

Pair wise alignment techniques : Database searching, Alphabets and complexity, Algorithm and programs, Comparing two sequences, sub-sequences, Identity and similarity, The Dot plot, Local and global similarity, different alignment techniques, Dynamic Programming, Pair wise database searching.

Multiple sequence alignment : Definition and Goal, The consensus, computational complexity, Manual methods, Simultaneous methods, Progressive methods, Databases of Multiple alignments and searching

Secondary database searching: Importance and need of secondary database searches, secondary database structure and building a sequence search protocol

Analysis packages : Analysis package structure, commercial databases, commercial software, comprehensive packages, packages specializing in DNA analysis, Intranet Packages, Internet Packages.

Text Books:

1. Introduction to Bioinformatics, T K Attwood & D J Parry-Smith Addison Wesley Longman.
2. Bioinformatics- A Beginner's Guide, Jean-Michel Claverie, Cerdric Notredame WILEY dreamlech India Pvt. Ltd

Reference Books:

1. Introduction to Bioinformatics, Arthur M.Lesk, OXFORD publishers (Indian Edition)

DATABASE MANAGEMENT SYSTEMS LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To introduce to a commercial DBMS such as ORACLE.
2. To learn and practice SQL commands for schema creation, data manipulation.
3. To learn conceptual and physical database design based on a case study.
4. To apply database design stages by studying a case study.

Course Outcomes:

1. The student is exposed to a commercial RDBMS environment such as ORACLE.
2. The student will learn SQL commands for data definition and manipulation.
3. The student understands conceptual through physical data base design.
4. The student takes up a case study and applies the design steps.

SYLLABUS

Features of a commercial RDBMS package such as ORACLE/DB2, MS Access, MYSQL & Structured Query Language (SQL) used with the RDBMS.

I. Laboratory Exercises Should Include:

- a. Defining Schemas for Applications,
- b. Creation of Database,
- c. Writing SQL Queries,
- d. Retrieve Information from Database,
- e. Creating Views
- f. Creating Triggers
- g. Normalization up to Third Normal Form
- h. Use of Forms
- i. Report Writing
- j. Implementation of enforcing Integrity constraints.

II. Some sample applications are given below:

1. Accounting Package for Shops,
2. Database Manager for Magazine Agency or Newspaper Agency,
3. Ticket Booking for Performances,
4. Preparing Greeting Cards & Birthday Cards
5. Personal Accounts - Insurance, Loans, Mortgage Payments, Etc.,
6. Doctor's Diary & Billing System

7. Personal Bank Account
8. Class Marks Management
9. Hostel Accounting
10. Video Tape Library,
11. History of Cricket Scores,
12. Cable TV Transmission Program Manager,
13. Personal Library.
14. Sailors Database
15. Suppliers and Parts Database

Reference Book:

1. Database System Concepts; A. Silberschatz, H. Korth 5th Edition, McGraw-Hill

WEB TECHNOLOGIES LAB

Lab : 3 Periods

Exam : 3 Hrs.

Sessionals : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. Implements the basic tags in HTML.
2. Develop web pages using Templates and Bootstrap.
3. Build dynamic pages using Servlets and PHP.
4. Develop mini project using PHP and Mysql.

Course Outcomes:

Upon successful completion of the course, the students would be able to

1. Understand current and evolving Web languages for integrating media and user interaction in both front end and back end elements of a Web site
2. Create static web pages using HTML and CSS
3. Validate HTML FORM data using JavaScript at the client side
4. Create dynamic web pages using PHP and MySQL
5. To build XML applications with schema and style sheets that span multiple domains for use with legacy browsers
6. Install tomcat and run servlet with for authentication and Session Management

LIST OF EXPERIMENTS**Week-1:**

Design the following static web pages required for an online book store web site.

1) HOME PAGE:

The static home page must contain three **frames**.

Top frame : Logo and the college name and links to Home page, Login page, Registration page, Catalogue page and Cart page (the description of these pages will be given below).

Left frame : At least four links for navigation, which will display the catalogue of respective links.

For e.g.: When you click the link “CSE” the catalogue for CSE Books should be displayed in the Right frame.

Right frame: The *pages to the links in the left frame must be loaded here*. Initially this page contains description of the web site.

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
CSE ECE EEE CIVIL	Description of the Web Site			

Fig 1.1

2) LOGIN PAGE:

This page looks like below:

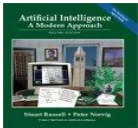
Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
CSE ECE EEE CIVIL	Login : Password: Submit Reset			

3) CATALOGUE PAGE:

The catalogue page should contain the details of all the books available in the web site in a table.

The details should contain the following:

1. Snap shot of Cover Page.
2. Author Name.
3. Publisher.
4. Price.
5. Add to cart button.

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
CSE		Book : XML Bible	\$ 40.5	
ECE		Author : Winston		
EEE		Publication : Wiely	\$ 63	
CIVIL		Book : AI		
		Author : S.Russel	\$ 35.5	
		Publication : Princeton hall		
		Book : Java 2		
		Author : Watson		
		Publication : BPB publications	\$ 50	
		Book : HTML in 24 hours		
		Author : Sam Peter		
		Publication : Sam publication		

Note: Week 2 contains the remaining pages and their description.

Week-2:

4) CART PAGE:

The cart page contains the details about the books which are added to the cart.

The cart page should look like this:

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
CSE	Book name	Price	Quantity	Amount
ECE				
EEE	Java 2	\$35.5	2	\$70
CIVIL	XML bible	\$40.5	1	\$40.5
			Total amount -	\$130.5

5) REGISTRATION PAGE:

Create a “registration form” with the following fields

- 1) Name (Text field)
- 2) Password (password field)
- 3) E-mail id (text field)
- 4) Phone number (text field)
- 5) Sex (radio button)
- 6) Date of birth (3 select boxes)
- 7) Languages known (check boxes – English, Telugu, Hindi, Tamil)
- 8) Address (text area)

WEEK 3:

VALIDATION:

Write *JavaScript* to validate the following fields of the above registration page.

1. Name (Name should contains alphabets and the length should not be less than 6 characters).
2. Password (Password should not be less than 6 characters length).
3. E-mail id (should not contain any invalid and must follow the standard pattern name@domain.com)
4. Phone number (Phone number should contain 10 digits only).

Note : You can also validate the login page with these parameters.

You can also use bootstrap validation also

WEEK 4:

Write a basic html by using following classes in bootstrap.css (External Style Sheet bootstrap.min.css)

1. Table
2. Button
3. Form
4. Dropdown

WEEK 5

Write an XML file which will display the Book information which includes the following:

- 1) Title of the book
- 2) Author Name
- 3) ISBN number
- 4) Publisher name
- 5) Edition
- 6) Price

Write a Document Type Definition (DTD) to validate the above XML file.

Display the XML file as follows. The contents should be displayed in a table. The header of the table should be in color GREY. Use XML schemas XSL and CSS for the above purpose

(Or)

Use Java Code to parse the xml

WEEK 6

- A) Installation of tomcat and Create Hello World message using servlet
- B) Read the user id and passwords entered in the Login form (week1) and authenticate (user id and passwords)using RequestDispatcher

WEEK 7

Demonstrate Session tracking in servlets using

1. Session API
2. Cookies

WEEK 8

Write an HTML page that has one input, which can take multi-line text and a submit button. Once the user clicks the submit button, it should show the number of characters ,lines and words in the text entered using an alert message. Words are separated with white space and lines are separated with new line character using PHP

WEEK 9

- a) PHP program to find number of page views (page Hit count using sessions)
- b) A web application that lists all cookies stored in the browser on clicking “list cookies” button, add cookies if necessary

WEEK 10

Using PHP, MYSQL develop a basic application that can

1. Register User
2. Display Users (For Admin login)
3. Update User Detail's (For Admin login)
4. Delete User (For Admin login)

Reference Books:

1. Internet , World Wide Web , How to program, Dietel , Nieto, PHI/PEA.
2. Web Tehnologies, Godbole, kahate, 2nd Ed., TMH

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

ADVANCED CODING
(Common for CSE & IT)
(MOOCS-I)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS**UNIT I Review Coding essentials and modular programming**

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

**IOS APPLICATION DEVELOPMENT
(MOOCS-I)**

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objective:

In this course student will able to importance of swift code and evaluation of swift from Objective C and they will know the basic syntax of swift and they can build basic IOS application

Course Outcomes:

Upon completion of this course, able to:

1. Able to know the principles of Swift
2. They can apply the Collections in real – world scenarios
3. Students can understand the features of Swift
4. They can understand the memory management of IOs application
5. Able to know the UI kit in swift

SYLLABUS

Industry Oriented Applications on following Topics

Introduction to Swift

Swift vs. Objective-C, Swift language principles, Variables, Types, and Control Flow, Variables and constants, Built-in types ,Conditional statements

Collections

Arrays, Dictionaries, Sets, Functions and Closures, Basic functions, Higher-order functions, nested functions, Closures

Classes and Structures

The Swift type system, Properties,Initializers,Methods, Property observers, Access modifiers, Inheritance and polymorphism, Structures

Mini project in Ios Application using Xcode

**RUBY
(MOOCS-I)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course Objective:

The objectives of this course include the following:

1. Experience learning a programming language "on your own" as is commonly the case in industry.
2. Understand the syntax and semantics of the Ruby language.
3. Understand how to develop and implement various types of programs in the Ruby language
4. Understand various forms of data representation and structures supported by the Ruby language.
5. Understand the appropriate applications of the Ruby language.
6. Recognize similarities and common characteristics of programming languages.

Course Outcome:

The student should develop or enhance skills in the following areas:

1. proficient programming in the Ruby language and programming in general
2. design and revision of Ruby scripts
3. debugging techniques appropriate for the Ruby language

SYLLABUS**UNIT-I****Introduction:**

Introduction to Ruby, What is Ruby? Features, Ruby gems, Help on ruby , Installation, variable Dynamic typing ,Declaring a variable ,Changing variable type , Converting the var value i.e to float,string,binary etc

UNIT-II**Ruby Methods and Arithmetic and Relational Operators**

What is method Working with methods, Declaring and Calling a method , Passing arguments to a method Explaining the arguments and syntax with example, Arithmetic Operators ,Addition Subtraction Multiplication Division Modulus, Exponent Precedence of operators Relational Operators.

UNIT-III :

Logical and other Operators and Control Statements

Logical operator And && Or || Not ! Parallel assignment Range operator Inclusive Operator(..) Exclusive operator (...),Control Statements in Ruby ,What are control statements Syntax for if statement if..else statement if..elsif statement Examples on it..

UNIT –IV :

For and each Looping Statements and while and until Looping Statements

for & each loops in Ruby, Meaning of the term “loop” *,Different kinds of loop ,Syntax of “for” loop*,Example implementation of “for” loop , Syntax of “each” loop , while & until loops in Ruby ,Usage of while loop with an example ,Usage of until loop with an example ,Usage of redo construct with an example

UNIT-V :

Object Oriented Concept in Ruby and Object Oriented Programming Methods

Object Oriented Concept in Ruby, Classes in Ruby , How to create objects ,Different ways of defining methods in Ruby ,Using ? and = to define meaningful methods, Methods in Ruby ,What are instance methods , class methods , accessor methods , Example implementation of each of the above and difference between them

References :

http://spoken-tutorial.org/tutorial-search/?search_foss=Ruby&search_language=English

WEB LINK:

<https://www.codecademy.com/learn/learn-ruby>

PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON**(MOOCS-I)**

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

This course is an introduction to programming and problem solving in Python. It does not assume any prior knowledge of programming. Using some motivating examples, the course quickly builds up basic concepts such as conditionals, loops, functions, lists, strings and tuples. It goes on to cover searching and sorting algorithms, dynamic programming and backtracking, as well as topics such as exception handling and using files. As far as data structures are concerned, the course covers Python dictionaries as well as classes and objects for defining user defined datatypes such as linked lists and binary search trees.

Course Outcomes :

1. Ability to apply object oriented concepts in programming
2. Ability to define, understand and differentiate different types of data types and apply them.
3. Ability to recognize various concepts of python dictionaries as well as classes and objects for defining user defined datatypes such as linked lists and binary search trees.

COURSE LAYOUT**Chapter 1**

Informal introduction to programming, algorithms and data structures via gcd

Downloading and installing Python

gcd in Python: variables, operations, control flow - assignments, condition-als, loops, functions

Chapter 2

Python: types, expressions, strings, lists, tuples

Python memory model: names, mutable and immutable values

List operations: slices etc

Binary search

Inductive function definitions: numerical and structural induction

Elementary inductive sorting: selection and insertion sort

In-place sorting

Chapter 3

Basic algorithmic analysis: input size, asymptotic complexity, $O()$ notation

Arrays vs lists

Merge sort

Quicksort

Stable sorting

Chapter 4

Dictionaries

More on Python functions: optional arguments, default values

Passing functions as arguments

Higher order functions on lists: map, iter, list comprehension

Chapter 5

Exception handling

Basic input/output

Handling files

String processing

Chapter 6

Backtracking: N Queens, recording all solutions

Scope in Python: local, global, nonlocal names

Nested functions

Data structures: stack, queue

Heaps

Chapter 7

Abstract datatypes

Classes and objects in Python

"Linked" lists: find, insert, delete

Binary search trees: find, insert, delete

Height-balanced binary search trees

Chapter 8

Efficient evaluation of recursive definitions: memoization

Dynamic programming: examples

Other programming languages: C and manual memory management

Other programming paradigms: functional programming

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs34

ANDROID APP DEVELOPMENT (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objective:

Android Application Development course is designed to quickly get you up to speed with writing apps for Android devices. The student will learn the basics of Android platform and get to understand the application lifecycle.

Course Outcome:

By the end of the course, student will be able to develop the android applications on their own, and work with the database to store data locally, and much more.

SYLLABUS

Unit 1: The Basics

Chapter1: 1.0 Intro to Android, 1.1 Create Your First Android App, 1.2 Layouts, Views and Resources, 1.3 Text and Scrolling Views.

Chapter2: 2.1 Activities and Intents, 2.2 Activity Lifecycle and Saving State, 2.3 Activities and Implicit Intents.

Unit 2: User Interface

1. **Chapter4:** 4.1 User Input Controls, 4.2 Menus, 4.3 Screen Navigation, 4.4 RecyclerView, fragments.
2. **Chapter5:** 5.1 Drawables, Themes and Styles, 5.2 Material design

Unit 3: Background tasks

1. **Chapter7:** 7.1 AsyncTask and AsyncTaskLoader, 7.2 Connecting to the Internet, 7.3 Broadcast receivers, 7.4 Services.
2. **Chapter8:** 8.1 Notifications, 8.2 Alarm managers, 8.3 Transferring data efficiently.

Unit 4: Data -- saving, retrieving, and loading

1. **Chapter9:** 9.0 Concepts: Overview to storing data, 9.1 Shared preferences, 9.2 App settings.
2. **Chapter10:** 10.1 SQLite primer, 10.2 store data using SQLite database.
3. **Chapter11:** 11.1 ContentProviders.

Unit 5: What's Next?

1. **Chapter13:** 13.1 Permissions, performance and security, Espresso, media playback.
2. **Chapter14:** 14.1 Firebase and AdMob, Firebase cloud messaging, places.
3. **Chapter15:** 15.1 Publish your app

WEB LINK:

1. <https://developer.android.com/develop/index.html>
2. <https://www.tutotrialspoint.com/>

**BLACKCHAIN ARCHITECTURE DESIGN AND USECASE
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objective:

The widespread popularity of digital cryptocurrencies has led the foundation of Blockchain, which is fundamentally a public digital ledger to share information in a trustworthy and secure way. The concept and applications of Blockchain have now spread from cryptocurrencies to various other domains, including business process management, smart contracts, IoT and so on. This course is a joint venture from academia and industry, where the target is to cover both the conceptual as well as application aspects of Blockchain. This includes the fundamental design and architectural primitives of Blockchain, the system and the security aspects, along with various use cases from different application domains.

Course Outcomes:

- 1.Student will able to understand what is Blockchain.
- 2.Student will able to undertstand about Bitcoin and how bitcoin and blockchain are related.
- 3.Student will able to understand consensus in Bitcoin.
- 4.Student will able to apply permissioned Blockchain.
5. Student will understand and learn to apply Blockchain in Government.
6. Student will understand and learn to apply Blockchain security.
7. Student will understand and learn to apply Blockchain in Research aspects,science and ecosystem..

Chapter	Lecture	Topic	Concepts
1.	1	Introduction to Blockchain – I (Basics)	What is Blockchain Public Ledgers Blockchain as public ledgers
	2	Introduction to Blockchain – II (History)	Bitcoin Blockchain 2.0 Smart Contracts
	3	Introduction to Blockchain – III (Architecture)	Block in a Blockchain Transactions Distributed Consensus
	4	Introduction to Blockchain – IV (Conceptualization)	The Chain and the Longest Chain Cryptocurrency to Blockchain 2.0 Permissioned Model of Blockchain
	5	Basic Crypto Primitives – I	Cryptographic Hash Function Properties of a hash function Hash pointer and Merkle tree
2.	6	Basic Crypto Primitives – II	Digital Signature Public Key Cryptography A basic cryptocurrency
	7	Bitcoin Basics – I	Creation of coins Payments and double spending FORTH – the precursor for Bitcoin scripting
	8	Bitcoin Basics – II	1) Bitcoin Scripts 2) Bitcoin P2P Network
	9	Bitcoin Basics – III	1) Transaction in Bitcoin Network 2) Block Mining 3) Block propagation and block relay
	10	Distributed Consensus	1) Why Consensus 2) Distributed consensus in open environments 3) Consensus in a Bitcoin network
3.	11	Consensus in Bitcoin – I (The Basics)	1) Bitcoin Consensus 2) Proof of Work (PoW) – basic introduction 3) Hashcash PoW
	12	Consensus in Bitcoin – II (PoW and Beyond)	1) Bitcoin PoW 2) Attacks on PoW and the monopoly problem 3) Proof of Stake, Proof of Burn and Proof of Elapsed Time
	13	Consensus in Bitcoin – III (The Miners)	1) The life of a Bitcoin Miner 2) Mining Difficulty 3) Mining Pool
	14	Permissioned Blockchain – I (Basics)	1) Permissioned model and use cases 2) Design issues for Permissioned blockchains 3) Execute contracts 4) State machine replication
	15	Permissioned Blockchain – II (Consensus)	1) Consensus models for permissioned blockchain 2) Distributed consensus in closed environment 3) Paxos

Chapter	Lecture	Topic	Concepts
4.	16	Permissioned Blockchain – III (RAFT Consensus)	1) RAFT Consensus 2) Byzantine general problem
	17	Permissioned Blockchain – IV (Byzantine General Problem)	1) Byzantine fault tolerant system 2) Lamport-Shostak-Pease BFT Algorithm
	18	Permissioned Blockchain – V (Practical Byzantine Fault Tolerance)	1) BFT over Asynchronous systems 2) Practical Byzantine Fault Tolerance 3) Three phase commit 4) View Change
	19	Blockchain for Enterprise - Overview	1) Concepts and benefits of block chain for enterprise 2) The Hyperledger Project
	20	Blockchain Components and Concepts	1) Actors in a Blockchain 2) Components in Blockchain design 3) Ledger in Blockchain
5.	21	Hyperledger Fabric – Transaction Flow	1) Fabric Architecture 2) Transaction flow in Fabric
	22	Hyperledger Fabric Details	1) Ordering Services 2) Channels in Fabric 3) Fabric Peer and Certificate Authority
	23	Fabric – Membership and Identity Management	1) Organization and Consortium Network 2) Membership Service Provide 3) Transaction Signing
	24	Hyperledger Fabric Network Setup	1) Steps for network setup 2) Endorsement policies
	25	Fabric Demo on IBM Blockchain Cloud – I	1) Setup Blockchain networks 2) Experience blockchain network as different organizations 3) Deploy a simple application on IBM cloud
6.	26	Fabric Demo on IBM Blockchain Cloud – II	1) Deploy a simple application on IBM Cloud 2) Marbles (asset transfer) 3) Example smart contract code, client SDK code 4) Perform blockchain transactions using a cool UI!
	27	Fabric Demo, deploy from scratch – III	1) Setup a Fabric network on your laptop or VM 2) Install and instantiate chaincode 3) Run application on the network you created
	28	Hyperledger Composer – Application Development	1) Goals of Hyperledger Composer 2) Key concepts for the business service provide 3) Key development concepts – Model files, Access control lists, Transaction processors, business network definition
	29	Hyperledger Composer – Network Administration	1) Key concepts for administrators 2) How composer maps to Fabric chaincode
	30	Blockchain Use Cases	1) Sample use cases by Industry 2) Business Problems and Participants 3) Communities in Blockchain network

Chapter	Lecture	Topic	Concepts
7.	31	Blockchain in Financial Service – I (Payments and Secure Trading)	1) Cross border payments 2) Stellar and Ripple protocols 3) Project Ubin
	32	Blockchain in Financial Service – II (Compliance and Mortgage)	1) Know Your Customer (KYC) 2) Privacy Consents 3) Mortgage over Blockchain
	33	Blockchain in Financial Service – III (Financial Trade)	1) Blockchain enabled Trade 2) We.Trade – Trade Finance Network 3) Supply Chain Financing
	34	Revolutionizing Global Trade	1) Blockchain for Trade Logistics 2) Global Trade Digitization 3) Blockchain for Container Management
	35	Blockchain in Supply Chain – I	1) Food Safety and Food Traceability 2) Supply Chain Orchestration
8.	36	Blockchain in Supply Chain – II	1) Everledger 2) The Diamond Lifecycle 3) Addressing Supply Chain Fraud through Blockchain
	37	Blockchain in Other Industries	1) Blockchain in Healthcare 2) Blockchain in Energy Markets 3) Blockchain in Media
	38	Blockchain in Government – I (Advantages)	1) Blockchain and Government 2) Preventing Cyber Crime through blockchain 3) Government Use-cases
	39	Blockchain in Government – II (Use Cases)	1) Auditing and Compliance 2) Blockchain for Defense 3) e-Estonia: A Case Study
	40	Blockchain in Government – III (Digital Identity)	1) Digital Identity and Single Sign On (SSO) 2) Principles of Digital Identity Management 3) Why Blockchain
9.	41	Blockchain in Government – IV (Hyperledger Indy)	1) Indy for Digital Identity Management 2) How Indy Works
	42	Blockchain in Government – V (Tax Payments and Land Registry Records)	1) Blockchain for Tax Payments 2) Blockchain for Managing Land Registry Records
	43	Blockchain Security – I (Overview)	1) Security properties 2) Security considerations for Blockchain 3) Intel SGX
	44	Blockchain Security – II (Membership and Access control in Fabric)	1) Identities and Policies 2) Membership and Access Control 3) Blockchain Crypto Service Providers
	45	Blockchain Security – III (Privacy in Fabric)	1) Privacy in a Blockchain System 2) Privacy through Fabric Channels 3) Smart Contract Confidentiality

Chapter	Lecture	Topic	Concepts
10.	46	Blockchain Security – III (Fabric SideDB)	1) SideDB Motivation 2) SideDB Overview
	47	Research Aspects – I (Consensus Scalability)	1) PoW vs BFT Consensus 2) Consensus Finality 3) Consensus Scalability
	48	Research Aspects – II (Bitcoin-NG)	1) Fairness and Scalability in Nakamoto Consensus 2) Bitcoin-NG: Working Principles 3) Key Blocks and Microblocks
	49	Research Aspects -III (Collective Signing)	1) Authority and Digital Signature 2) Collective Signing (CoSi) 3) Shnorr Multisignature and BLS Signatures
	50	Research Aspects – IV (Byzcoin)	1) Strong non-probabilistic consistency 2) BFT over Bitcoin – increasing scalability 3) Byzcoin Design and Performance
11.	51	Research Aspects – V (Algorand)	1) Strong Synchrony vs Weak Synchrony 2) Avoiding Forks 3) Transaction Neutrality and Frictionless Evolution
	52	Research Aspects – VI (Cross Fault Tolerance)	1) Asynchronous networks as network fault 2) Cross fault Tolerant (XFT) architecture 3) XPaxos
	53	Research Aspects – VII (Secured Multi-Party Computation)	1) Multi-Party Computation (MPC) 2) Fairness in MPC 3) MPC over Blockchain – ensuring fairness
	54	Blockchain for Science – I (Blockchain for Big Data)	1) Big Data and Big Network 2) Why Blockchain for Big data – application aspects 3) BigChainDB – The Blockchain Database
	55	Blockchain for Science – II (Blockchain and AI)	1) Data analysis over Blockchain 2) Logic over Blockchain network 3) Inferring Decisions through AI
12.	56	Comparing Ecosystems – Ethereum	1) Architecture and concepts 2) Smart contracts and EVM 3) Additional/upcoming capabilities
	57	Comparing Ecosystems – Ethereum development tools and Quorum	1) Ethereum Development Tools 2) Motivation and concepts 3) Architecture 4) Transaction processing and consensus
	58	Comparing Ecosystems – Corda Part 1& Comparing Ecosystems – Corda Part 2	1) Key features, Notaries and Oracles 2) Transactions, Transaction Flows 3) Corda ledger and smart contracts, Consensus

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs47/course

TESTING TOOLS (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objective: Testing tools course is designed to learn the basics of Testing Tools and get to understand the Testing process.

Course Outcomes: By the end of the course, student will be able to know various testing tools like HP-Quality Center-11.00, Performance Test Automation with LOADRUNNER 11.00 & Browser Automation Testing Tool SELENIUM-1.10.0

SYLLABUS

UNIT-I : Testing Introduction

Introduction, Importance of software systems, Common problems in software development, **SDLC Implementation:** Feasibility study/Requirement Gathering, Analysis, Designing, Coding, Testing, Delivery and Maintenance, **Software Development Models:** Waterfall Model, Incremental Model, Prototype Model/Use and Through Model, Spiral Model, V-Model, W-Model, Agile Testing Implementation, Agile-SCRUM, **Kinds of Testing:** Un-Conventional Testing, Conventional Testing, **Methods of Testing :** Black Box Testing, White Box Testing, Gray Box Testing. **Static Testing:** Reviews, Inspections, Audit, Walkthrough. **Dynamic Testing:** Unit Testing, Module Testing, Integration Testing, System Testing, User Acceptance Testing

UNIT-II: HP-Quality Center-11.00

Overview on Test Management, Quality Center – Introduction, Need of Test-Management Tool, Module (TestDirector Project, Site Administration, Customization), Domain/Project Fundamentals, How to Get Started

Architecture of TD/QC Tool: Site Administrator, Create Domain, Create Project, Create Users, Assign Users to Project, Monitor Connections

Customization of Quality Center Release and Cycle creation Test Requirements: Example of a test requirement, Importance of tracing and tracking Requirements, Reviewing and building a, Requirements structure, Entering requirements manually

Test Cases Creation and management: Review of an existing test case, Parameters, Building a test case structure, Creating manual test cases, Requirements coverage

Test Sets and Test Execution: Creating folders and test sets, Defining test execution flow, Setting test set properties, Manual test execution, Logging defects during manual testing, Automated test execution, Adding test hosts, Running a test set, Setting run times

Defect Tracking: Reporting defects, Searching for similar defects, Using grid filters, Deleting defects

Reporting and Analysis: Analysis menu graphs and reports, Creating editable reports with the advanced, Reporting

Bug Tracking Tools: BugZilla, Mantis, PRTracker, VSS

Automation Life Cycle Implementation Using HP-QuickTest Professional-11.00

Introduction to QuickTest: Overview of Add-In Manager and QuickTest, QuickTest Window and Tools, QuickTest Commands, Tools in QuickTest

UNIT-III : Performance Test Automation with LOADRUNNER 11.00

Introduction, Overview of LoadRunner, The need of performance testing, When load testing comes into process, Performance Testing, Stress Testing, LoadRunner Architecture, LoadTest Process

Performance Test Process: Manual Load Testing, Automation Load Testing, Performance Test Approach, Study system under test, Create Vuser Script, Execute Tests, Analyze Test Results

Load Runner Components: Virtual user generator, Controller, Analysis

Load testing process: Building Tests, Running Load Tests, Analyze the results, Load Runner Controller, Publish Reports

UNIT-IV: Browser Automation Testing Tool SELENIUM-1.10.0

Introduction to Selenium, What is Selenium, Use of Selenium, When Selenium can be useful in testing, Features of Selenium, Differences between Selenium and QTP

Selenium Components: Selenium IDE, Selenium RC, Selenium Grid

WEB LINKS:

<https://www.tutorialspoint.com/qc/index.htm>

<https://www.udemy.com/starthpalm/>

<https://www.udemy.com/loadrunner/>

MACHINE LEARNING (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

This course provides a concise introduction to the fundamental concepts in machine learning and popular machine learning algorithms. We will cover the standard and most popular supervised learning algorithms including linear regression, logistic regression, decision trees, k-nearest neighbor, an introduction to Bayesian learning and the naïve Bayes algorithm, support vector machines and kernels and neural networks with an introduction to Deep Learning. We will also cover the basic clustering algorithms. Feature reduction methods will also be discussed. We will introduce the basics of computational learning theory. In the course we will discuss various issues related to the application of machine learning algorithms. We will discuss hypothesis space, over fitting, bias and variance, tradeoffs between representational power and learn ability, evaluation strategies and cross-validation. The course will be accompanied by hands-on problem solving with programming in Python and some tutorial sessions.

Course Outcomes:

1. Have a good understanding of the fundamental issues and challenges of machine learning: data, model selection, model complexity, etc.
2. Have an understanding of the strengths and weaknesses of many popular machine learning approaches.
3. Appreciate the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
4. Be able to design and implement various machine learning algorithms in a range of real-world applications.

SYLLABUS

Chapter1:

Introduction: Basic definitions, types of learning, hypothesis space and inductive bias, evaluation, cross-validation.

Chapter2:

Linear regression, Decision trees, over fitting.

Chapter3:

Instance based learning, Feature reduction, Collaborative filtering based recommendation.

Chapter4:

Probability and Bayes learning.

Chapter5:

Logistic Regression, Support Vector Machine, Kernel function and Kernel SVM.

Chapter6:

Neural network: Perception, multilayer network, back propagation, introduction to deep neural network.

Chapter7:

Computational learning theory, PAC learning model, Sample complexity, VC Dimension, Ensemble learning.

Chapter8:

Clustering: k-means, adaptive hierarchical clustering, Gaussian mixture model.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs40/

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)
III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System
INFORMATION TECHNOLOGY
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 3201	Data Ware Housing & Data Mining	4	3	1	--	4	30	70	100
B16 IT 3202	Object Oriented Software Engineering	4	3	1	--	4	30	70	100
B16 IT 3203	Design and Analysis of Algorithms	4	3	1	--	4	30	70	100
B16 IT 3204	Compiler Design	4	3	1	--	4	30	70	100
B16 IT 3205	Cryptography And Network Security	4	3	1	--	4	30	70	100
#ELE-II	ELECTIVE-II	4	3	1	--	4	30	70	100
B16 IT 3211	Software Engineering And Mini Project Lab	2	--	--	3	3	50	50	100
B16 IT 3212	Computer Graphics and Multimedia Lab	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantative Aptitude-II	2	5	--	--	5	100	--	100
#M-III	MOOCS-III	2	4	--	--	4	100	--	100
#M-IV	MOOCS-IV	2	4	--	--	4	100	--	100
Total		34	31	6	6	43	580	520	1100

#ELE-II	B16 IT 3206	Image Processing
	B16 IT 3207	Distributed Database Systems
	B16 IT 3208	Computer Graphics
	B16 IT 3209	Mobile Computing
	B16 IT 3210	Soft Computing And Neural Networks.
#M-III	B16 ENG 3205	Competitive Coding
	B16 IT 3213A	Amazon Web Services
	B16 IT 3213B	ASP.NET
	B16 IT 3213C	Routing & Switching
#M-IV	B16 IT 3214A	Data Science using Advanced Python
	B16 IT 3214B	Angular JS
	B16 IT 3214C	C#.NET and VB.NET
	B16 IT 3214D	MAT LAB

DATAWAREHOUSING & DATA MINING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives :

1. To understand the evolution of data warehousing and data mining systems
2. To understand extracting, cleaning and transformation of data into a warehouse.
3. To learn the principles of statistics, information theory, machine learning and other areas AI and implementation of data mining techniques.
4. To understand pattern mining using classification and clustering methods.

Course Outcomes:

1. The student understands the differences between OLTP and OLAP.
2. The student learns how data cube technology supports summarization and querying high dimensional data.
3. The student is introduced to similarity, distance, information gain and other performance and error metrics used for evaluation of mining results.
4. The student is introduced to various approaches to association rule mining , supervised and unsupervised learning and the corresponding classification and clustering approaches involving decision trees, Bayesian approaches, model based and agglomerative approaches.

SYLLABUS

Introduction to Data Mining: Evolution of IT into DBMS, Motivation and importance of Data Warehousing and Data Mining, Kinds of Patterns, Technologies, Applications, Major Issues in Data Mining, Data Objects and Attributes Types, Statistical Descriptions of Data, Data Visualization, Estimating Data Similarity and Dissimilarity,

Data pre-processing : Quality data, Data Cleaning, Data Integration, Data Reduction, Data Transformation, Discretization and Concept Hierarchy Generation.

Data Warehouse and OLAP Technology: Basic Concepts of Data warehouse, Data Modeling using Cubes and OLAP, DWH Design and usage, Implementation using Data Cubes and OLAPs, Data Generalization with AOI.

Data Cube Technology: Preliminary Concepts of Data Cube Computation, Data Cube Computation Methods: Multi-way Array Aggregation for Full Cube, Multi-dimensional Data Analysis in cube space,

Mining Frequent Patterns Based on Associations and Correlations: Basic Concepts, Frequent Itemset Mining Methods: Apriori Algorithm, Association Rule Generation, Improvements to Apriori, FP- Growth Approach, Mining Closed and Max Patterns, Pattern Evaluation Methods, Association mining in multi-level, multi-dimensional space

Classification & Prediction: Basic Concepts, Decision Tree Induction, Bayes Classification, Rule- Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy, Classification by Back Propagation, Associative Classification, K-nearest neighbor classifier.

Cluster Analysis: Basic Concepts and issues in clustering, Types of Data in Cluster Analysis, Partitioning Methods, Hierarchical Methods, Density Based Methods, Grid Based Methods, Evaluation of Clustering Solutions

Text Book:

1. Data Mining- Concepts and Techniques by Jiawei Han, Micheline Kamber and Jian Pei – Morgan Kaufmann publishers ---3rd edition

Reference Books:

1. Introduction to Data Mining, Adriaan, Addison Wesley Publication
2. Data Mining Techniques, A.K.Pujari, University Press Data mining concepts by Tan, Steinbech, and Vipin Kumar - Pearson Edu publishers

OBJECT ORIENTED SOFTWARE ENGINEERING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives:

1. To explain the importance of OOSE in Software development.
2. To explain the students the importance of Requirements Engineering.
3. To explain the role of UML and Testing in Software Development.
4. To explain the entire Software Development Process with aid of case studies.

Course Outcomes:

1. Ability to define a problem and perform Requirements Engineering.
2. Ability to draw UML diagrams for the requirements gathered.
3. Ability to implement the designed problem in Object Oriented Programming Language.
4. Test whether all the requirements specified have been achieved or not.

SYLLABUS**Introduction to Object Oriented Software Engineering & Requirements Engineering:**

Nature of the Software, Types of Software, Software Engineering Projects, Software Engineering Activities, Software Quality, Introduction to Object Orientation, Software Process Models- Waterfall Model, Opportunistic Model, Phased Released Model, Spiral Model, Evolutionary Model, Concurrent Engineering Model, Domain Analysis, Problem Definition and Scope, Requirements Definition, Types of Requirements, Techniques for Gathering and Analyzing Requirements.

Unified Modeling Language & Use Case Modeling: Modeling Concepts, Types of UML Diagrams with Examples; User-Centered Design, Characteristics of Users, Developing Use-Case Models of Systems, Use-Case Diagram, Use- Case Descriptions, Usability Principles, Interaction and Behavioral Diagrams: Interaction Diagrams, State Diagrams, Case Study.

Class Design and Class Diagrams: Activity Diagrams, Essentials of UML Class Diagrams, Associations and Multiplicity, Other Relationships, Generalization, Instance Diagrams, Advanced Features of Class Diagrams, Component and Deployment Diagrams, Case Study.

Software Design and Architecture: Process of Design, Principles Leading to Good Design, Techniques for Making Good Design Decisions, Good Design Document; Pattern Introduction, Design Patterns: Abstraction-Occurrence Pattern, General Hierarchical Pattern, Play-Role Pattern, Singleton Pattern, Observer Pattern, Delegation Pattern, Adaptor Pattern, Façade Pattern,

Immutable Pattern, Read-Only Interface Pattern and The Proxy Pattern; Software Architecture Contents of Architecture Model, Architectural Patterns: Multilayer, Client-Server, Broker, Transaction Processing, Pipe & Filter and MVC Architectural Patterns.

Software Testing & Software Process Management: Overview of Testing, Testing Concepts, Testing Activities, Testing Strategies, Unit Testing, Integration Testing, Function Testing, Structural Testing, Class Based Testing Strategies, Use Case/Scenario Based Testing, Regression Testing, Performance Testing, System Testing, Acceptance Testing, Installation Testing, OO Test Design Issues, Test Case Design, Quality Assurance, Root Cause Analysis, Post-Mortem Analysis, Introduction to Software Project Management, Rationale Management, Configuration Management, Activities of Software Project Management, Structure of Project Plan, Software Cost Estimation, Project Scheduling.

CASE STUDY:

1. Simple Chat Instant Messaging System
2. GPS Based Automobile Navigation System
3. Waste Management Inspection Tracking System (WMITS)
4. Geographical Information System

Text Books:

1. Object-Oriented Software Engineering Practical software development using UML and Java by Timothy C. Lethbridge & Robert, Langanieri McGraw-Hill.
2. Object-Oriented Software Engineering: Using UML, Patterns and Java, Bernd Bruegge and Allen H. Dutoit, 2nd Edition, Pearson Education Asia.

Reference Books:

1. Software Engineering: A Practitioner's Approach, Roger S Pressman.
2. A Practical Guide to Testing Object-Oriented Software, John D. McGregor; David A. Sykes, Addison-Wesley Professional.
3. Software Engineering, K.K. Agarwal, New Age Publications 2008

DESIGN AND ANALYSIS OF ALGORITHMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

On completing this course student will be able to

1. Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions
2. Analyze the asymptotic performance of algorithms.
3. Demonstrate a familiarity with major algorithms and data structures.
4. Synthesize efficient algorithms in common engineering design situations.

Course Outcomes :

1. Students will be able to Analyze the algorithms using asymptotic analysis.
2. Student will be able to understand, apply and analyze Divide-and-Conquer technique on computer science problems.
3. Student will be able to understand, apply and analyze Greedy technique on computer science problems.
4. Student will be able to understand, apply and analyze Dynamic Programming on computer science problems.
5. Student will be able to understand, apply and analyze Basic Traversal and Search techniques and Backtracking on computer science problems.
6. Student will be able to understand, apply and analyze Branch-and-Bound

SYLLABUS

Introduction: Algorithm, Algorithm specification, Performance Analysis.

Divide and Conquer: The general method -- Binary search -- finding maximum and minimum - Merge sort -- Quick sort -- Selection – Strassen’s Matrix multiplication – Convex Hull

The Greedy Method: The general method – Knapsack problem -- Job sequencing with deadlines -- Optimal storage on tapes -- minimum cost spanning trees -- Optimal Merge Patterns -- single source shortest paths.

Dynamic Programming: The general method -- Multistage graphs -- all pairs shortest paths -- optimal binary search trees -- reliability design -- the traveling sales person problem.

Basic Traversal and Search techniques: Techniques for Binary trees – Techniques for Graphs – Connected components and spanning trees -- Bi-connected components and depth first search.

Back Tracking: The General Method -- Eight Queens problem -- Sum of subsets -- Graph coloring -- Hamiltonian cycle.

Branch and Bound: The Method--Least cost (LC) Search-- The 15-Puzzle: an Example—Control Abstraction for LC-Search-- 0/1Knapsack Problem--Traveling Salesperson

Text Books:

1. Fundamentals of Computer Algorithms by Ellis Horowitz, S. Sahni et.al. Galgotia Pub. 2001.
2. The Design and Analysis of Computer Algorithms by Aho, Hopcroft, Ullman.

Reference Books:

1. Computer Algorithms: Introduction to Design and Analysis by Sara Baase, Allen Van Gelder, Pearson Education.
2. Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, Pearson Education.

COMPILER DESIGN

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To introduce the major concept areas of language translation and compiler design.
2. To enrich the knowledge in various phases of compiler and its use, code optimization techniques, machine code generation, and use of symbol table.
3. To extend the knowledge of parser by parsing LL parser and LR parser.

Course Outcomes:

1. To apply the knowledge of lex tool & yacc tool to develop a scanner & parser.
2. To design & conduct experiments for Intermediate Code Generation in compiler.
3. To design & implement a software system for backend of the compiler.
4. To learn the new code optimization techniques to improve the performance of a program in terms of speed & space.
5. To acquire the knowledge of modern compiler & its features.
6. To learn & use the new tools and technologies used for designing a compiler

SYLLABUS**Introduction to Compilers:**

Translators-Compilation and Interpretation-Language processors-The Phases of Compiler-Errors encountered in Different Phases-The Grouping of Phases-Compiler Construction Tools. Programming language basics.

Lexical Analysis

Need and Role of Lexical Analyzer-Lexical Errors-Expressing Tokens by Regular Expressions-Converting Regular Expression to DFA- Minimization of DFA-Language for Specifying Lexical Analyzers-LEX-Design of Lexical Analyzer for a sample language.

Syntax Analysis:

Need and Role of the Parser-Context Free Grammars -Top Down Parsing -General Strategies-Recursive Descent Parser Predictive Parser-LL(1) Parser-Shift Reduce Parser-LR Parser-LR (0)Item-Construction of SLR Parsing Table -Introduction to LALR Parser - Error Handling and Recovery in Syntax Analyzer-YACC-Design of a syntax Analyzer for a Sample Language .

Syntax Directed Translation & Run Time Environment:

Syntax directed Definitions-Construction of Syntax Tree-Bottom-up Evaluation of S-Attribute Definitions- Design of predictive translator - Type Systems-Specification of a simple type checker-Equivalence of Type Expressions-Type Conversions. RUN-TIME ENVIRONMENT: Source Language Issues-Storage Organization-Storage Allocation-Parameter Passing-Symbol Tables-Dynamic Storage Allocation -Storage Allocation in FORTRAN.

Code Optimization And Code Generation

Principal Sources of Optimization-DAG- Optimization of Basic Blocks-Global Data Flow Analysis-Efficient Data Flow Algorithms-Issues in Design of a Code Generator - A Simple Code Generator Algorithm.

Text Book

1. Principles of Compiler Design by Aho, D. Ullman

Reference Books

1. Compiler Construction by Kenneth. C. Louden, Vikas Pub. House.

CRYPTOGRAPHY & NETWORK SECURITY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Introduction of the issues in network security- its need and importance, taxonomy and terminology.
2. Discussion of various cryptographic techniques.
3. Exploration of different types of security threats and remedies.
4. Understanding of Internet security protocols and standards.

Course Outcomes: At the end of the Course, Student will be able to:

1. Realize the need and importance of network and data security in the Internet and in the distributed environments.
2. To be familiar with different means of Authentication mechanisms.
3. Identify the different types of network security issues and their remedies.
4. Application of various cryptographic tools and techniques in different contexts.
5. To be familiar with some internet security protocols and standards.

SYLLABUS

Overview: Computer Security Concepts, Threats, Attacks, and Assets, A Security Architecture for Open Systems, Cryptographic Tools: Confidentiality with Symmetric Encryption, Message Authentication and Hash Functions, Public-Key Encryption, Digital Signatures and Key Management, Random and Pseudorandom Numbers. User Authentication: Means of Authentication, Password-Based Authentication, Token-Based Authentication, Biometric Authentication, Remote User Authentication, Security Issues for User Authentication.

Access Control: Access Control Principles, Subjects, Objects, and Access Rights, Discretionary Access Control, Role-Based Access Control, Database Security: The Need for Database Security, Database Management Systems, Relational Databases, Database Access Control, Inference, Statistical Databases, Database Encryption, Cloud Security.

Malicious Software: Types of Malicious Software (Malware), Propagation—Infected Content—Viruses, Propagation—Vulnerability Exploit—Worms, Propagation— Social Engineering—SPAM E-mail, Trojans, Payload—System Corruption, Payload—Attack Agent—Zombie, Bots, Payload—Information Theft—Key loggers, Phishing, Spyware, Payload—Stealth—Backdoors, Root kits, Countermeasures. Denial-of-Service Attacks: Denial-of-Service Attacks, Flooding Attacks, Distributed Denial-of-Service Attacks, Application-Based Bandwidth Attacks, Reflector and Amplifier Attacks, Defenses Against Denial-of-Service Attacks, Responding to a Denial-of-Service Attack.

Intrusion Detection: Intruders, Intrusion Detection, Host-Based Intrusion Detection, Distributed Host-Based Intrusion Detection, Network-Based Intrusion Detection, Intrusion Detection Exchange Format, Honeypots. Firewalls and Intrusion Prevention Systems: The Need for Firewalls, Firewall Characteristics, Types of Firewalls, Firewall Basing, Firewall Location and Configurations, Intrusion Prevention Systems.

Symmetric Encryption and Message Confidentiality: Symmetric Encryption Principles, Data Encryption Standard, Advanced Encryption Standard, Stream Ciphers and RC4, Cipher Block Modes of Operation, Location of Symmetric Encryption Devices, Key Distribution. Public-Key Cryptography and Message Authentication: Secure Hash Function, HMAC, The RSA Public-Key Encryption Algorithm, Diffie-Hellman Key Exchange Algorithm.

Internet Security Protocols and Standards: Secure E-mail and S/MIME, Domain Keys Identified Mail, Secure Socket Layer (SSL) and Transport Layer Security (TLS), HTTPS, IPv4 and IPv6 Security. Internet Authentication Applications: Kerberos, X.509, Public-Key Infrastructure, Federated Identity Management. Wireless Network Security: IEEE 802.11 Wireless LAN Overview, IEEE 802.11i Wireless LAN Security.

Text Book:

1. Computer Security - Principles and Practices 2nd Edition by William Stallings, Pearson Education, Inc.

Reference Books:

1. Cryptography and Network Security by William Stallings, Pearson Education Asia, New Delhi.
2. Network Security Essentials Applications and Standards, by William Stallings, Pearson Education Asia, New Delhi.

IMAGE PROCESSING (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives:

1. To explain fundamentals of Image processing concepts.
2. To provide mathematical foundation of image enhancement , image compression and image segmentation.
3. To explain the students about Morphology and its applications in image processing.
4. To explain various methods and techniques for image transformation.

Course outcomes

1. Ability to develop algorithms for fundamental concepts in Image processing.
2. Ability to perform image enhancement , image compression and image segmentation using various methods.
3. Ability to implement Image transformation techniques

SYLLABUS

Fundamentals of Image Processing: Image Acquisition, Image Model, Sampling, Quantization, Relationship Between Pixels, Distance Measures, Connectivity , Image Geometry, Photographic Film. Histogram: Definition, Decision Of Contrast Basing On Histogram, Operations Basing on Histograms Like Image Stretching, Image Sliding, Image Classification. Definition and Algorithm of Histogram Equalization.

Image Enhancement in Spatial Domain : Arithmetic and Logical Operations, Pixel or Point Operations, Size Operations; Smoothing Filters-Mean, Median, Mode Filters – Comparative Study.

Edge enhancement in spatial domain: Edge enhancement filters, Directorial Filters, Sobel, Laplacian, Robert, KIRSCH Homogeneity & DIFF filters, PREWITT Filter, Contrast based edge enhancement techniques, Comparative study, Low pass filters, High pass filters, Sharpening filters, Comparative study, Color fundamentals and color model

Image Compression: Run Length Encoding, modified run length encoding, Contour Coding, Huffman Code, Compression Due to Change in Domain, Compression Due to Quantization Compression at the Time of Image Transmission. Brief Discussion on:- Image Compression Standards.

Image Segmentation: Definition of segmentation, Characteristics of Segmentation, Detection of Discontinuities, Thresholding. Pixel Based Segmentation Method. Region Based Segmentation Methods, Segmentation by Pixel Aggregation, Segmentation by Sub Region Aggregation, Histogram Based Segmentation, Spilt and Merge Technique, Segmentation of moving objects

Morphology: Dilation, Erosion, Opening, Closing, Hit-And-Miss Transform, Thinning, Thickening, Skeletons , Pruning Extensions to Gray – Scale Images Application of Morphology in I.P

Image Transforms : A Detail Discussion On Fourier Transform, DFT,FFT, Properties of Fourier transform, WALSH Trans Form , WFT, HADAMARD Transform, DCT Image Enhancement in Frequency Domain: Design of Low Pass, High Pass, EDGE Enhancement, Smoothing Filters in Frequency Domain. Butter Worth Filter, Homomorphic Filters in Frequency Domain Advantages of Filters in Frequency Domain, Comparative Study of Filters in Frequency, Domain and Spatial Domain.

Text Book:

1. Digital Image Processing, Rafael C. Gonzalez And Richard E. Woods, Addison Wesley

Reference Books:

1. Fundamentals Of Electronic Image Processing By Arthyr – R – Weeks, Jr.(PHI).
2. Image Processing, Analysis, And Machine Vision By Milan Sonka Vaclan Halavac Roger Boyle, Vikas Publishing House.
3. Digital Image Processing, S. Jayaraman, S. Esakkirajan& T. Veera Kumar, TMH.
4. Fundamentals of Digital Image Processing, Chris Solomon, Tobi Breckon, Wiley-Blackwell

DISTRIBUTED DATABASE SYSTEMS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. This course will introduce principles and foundations of distributed databases.
2. This course will introduce basic architecture, design issues.
3. This course will introduce strategies of integrity control, query processing and optimization.
4. This course will introduce principles transactions, and concurrency control mechanisms.

Course Out comes :

1. Explain the techniques used for data fragmentation, replication, and allocation during the distributed database design process.
2. Evaluate simple strategies for executing a distributed query to select the strategy that minimizes the amount of data transfer.
3. Explain how the two-phase commit protocol is used to deal with committing a transaction that accesses databases stored on multiple nodes.
4. Describe distributed concurrency control based on the distinguished copy techniques and the voting methods

SYLLABUS

Features of Distributed versus Centralized Databases, Principles Of Distributed Databases , Levels Of Distribution Transparency, Reference Architecture for Distributed Databases , Types of Data Fragmentation, Integrity Constraints in Distributed Databases.

Translation of Global Queries to Fragment Queries, Equivalence Transformations for Queries, Transforming Global Queries into Fragment Queries, Distributed Grouping and Aggregate Function Evaluation, Parametric Queries.

Optimization of Access Strategies, A Framework for Query Optimization, Join Queries, General Queries.

The Management of Distributed Transactions, A Framework for Transaction Management, Supporting Atomicity of Distributed Transactions, Concurrency Control for Distributed Transactions, Architectural Aspects of Distributed Transactions.

Concurrency Control, Foundation of Distributed Concurrency Control, Distributed Deadlocks, Concurrency Control based on Timestamps, Optimistic Methods for Distributed Concurrency Control.

Reliability, Basic Concepts, Nonblocking Commitment Protocols, Reliability and concurrency Control, Determining a Consistent View of the Network, Detection and Resolution of Inconsistency, Checkpoints and Cold Restart, Distributed Database Administration, Catalog Management in Distributed Databases, Authorization and Protection

Architectural Issues, Alternative Client/Server Architectures, Cache Consistency Object Management, Object Identifier Management, Pointer Swizzling, Object Migration, Distributed Object Storage, Object Query Processing, Object Query Processor Architectures, Query Processing Issues, Query Execution , Transaction Management, Transaction Management in Object DBMSs , Transactions as Objects.

Database Integration, Scheme Translation, Scheme Integration, Query Processing Query Processing Layers in Distributed Multi-DBMSs, Query Optimization Issues. Transaction Management Transaction and Computation Model Multidatabase Concurrency Control, Multidatabase Recovery, Object Orientation And Interoperability Object Management Architecture CORBA and Database Interoperability Distributed Component Model COM/OLE and Database Interoperability, PUSH-Based Technologies

Text Books :

1. Distributed Database Principles & Systems, Stefano Ceri, Giuseppe Pelagatti McGraw-Hill

Reference Books:

1. Principles of Distributed Database Systems, M.Tamer Ozsu, Patrick Valduriez – Pearson Education.

COMPUTER GRAPHICS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Provides a comprehensive introduction to computer graphics with a foundation in Graphics Applications.
2. A thorough introduction to computer graphics techniques.
3. To give the basics of Geometric Transformations and projections.
4. To introduce three dimensional concepts and object representations with color models and basics of computer animation.

Course Outcomes:

1. The students will understand graphics principles and graphics hardware.
2. The students can demonstrate geometrical transformations.
3. The students can create interactive graphics applications and demonstrate computer graphics animation.

SYLLABUS

Introduction: Computer Graphics and their applications: Computer Aided Design, Computer Art, Entertainment, Education and Training, Graphical User Interfaces; Overview of Graphics systems: Video Display Devices, Raster Scan Systems, Random Scan Systems, Graphics Monitors And Workstations, Input Devices, Hard Copy Devices, Interactive Input Methods, Windows and Icons, Virtual Reality Environments, Graphics Software.

Output primitives : Points and Lines, , Line and Curve Attributes, Color and Gray scale levels, Antialiasing, Loading the Frame buffer, Line function, Line Drawing Algorithms, Circle Generating Algorithms, Ellipse Generating Algorithms, Pixel Addressing, Area Fill Attributes, Filled Area Primitives, Filled Area Functions, Cell Array, Character Generation, Character Attributes, Bundled Attributes, Curve Functions, Parallel Curve Algorithms.

Two Dimensional Transformations: Basic 2D Transformations, Matrix Representations, Homogeneous Coordinates, Composite Transformations, Other Transformations, Transformations between Coordinate Systems, Affine Transformations.

Three Dimensional Transformations & Projections: Translation, Rotation, Scaling, Other Transformations, Composite Transformations, 3D Transformation Functions, Modeling and Coordinate Transformations, Need for projections, Parallel & Perspective projections, General Projection Transformations.

Viewing Pipeline and Clipping operations : Viewing Pipeline ,Viewing Coordinates & Reference frames, Window-to-Viewport Coordinate Transformation, Two Dimensional Viewing Functions, , Three Dimensional Viewing, View Volumes, Clipping and its Operations, Types of clipping operations- Point Clipping,Line Clipping, Polygon Clipping,, Curve Clipping,, Text and Exterior Clipping.

Three Dimensional Concepts and Object representations: 3D display methods, 3D Graphics, Polygon Surfaces, Curved Lines and Surfaces, Quadratic Surfaces, Super Quadrics, Blobby Objects, Spline Representations, Bézier Curves and Surfaces, BSpline Curves and Surfaces.

Introduction to Multimedia: Multimedia and Hypermedia, Worldwideweb, Overview multimedia tools, Graphics and Image Data Representation.

Text Book:

1. Computer Graphics, Donald Hearn & M.Pauline Baker, Pearson Education ,New Delhi.
2. Fundamentals of Multimedia, Ze-Nian Li, Mark S. Drew, Pearson Education ,New Delhi.

Reference Books:

1. Procedural Elements for Computer Graphics, David F.Rogers, Tata McGrawHill Book Company, New Delhi, 2003.
2. Computer Graphics: Principles & Practice in C,J.D.Foley, S.KFeiner, A Van Dam F.H John Pearson Education,2004.
3. Computer Graphics using OpenGL,FranscisSHillJr,PearsonEducation,2004.
4. Computer Vision and Image Processing: A Practical Approach using CVIPtools, S. E. Umbaugh,, Prentice Hall, 1998

MOBILE COMPUTING (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To make the student understand the concept of mobile computing paradigm, Its novel applications and limitations.
2. To understand the typical mobile networking Infrastructure through a popular GSM protocol.
3. To understand the issues and solutions of various layers of mobile networks, namely MAC layer, Network Layer & Transport Layer.
4. To understand the database issues With mobile environments & data delivery models.
5. To understand the ad hoc networks and related concepts.
6. To understand the platforms and protocols used in mobile environment.

Course Outcomes:

1. Able to think and develop new mobile application.
2. Able to take any new technical issue related to s new paradigm and come up with a solution(s).
3. Able to develop new ad hoc network applications and/or/ algorithms/ protocols.
4. Able to understand & develop any existing or new protocol related to mobile environment.

SYLLABUS

Introduction: Mobile Communications, Mobile Computing – Paradigm. Promises/Novel Applications and impediments and Architecture; Mobile and Handheld Devices, Mobile and Handheld Devices. GSM — Services. System Architecture. Radio Interfaces, Protocols. Localization, Calling, Handover, Security, New Data Services, GPRS, CSHSD, DECT.

(Wireless) Medium Access Control (MAC): Motivation for a specialized MAC (Hidden and exposed terminals. Near and far terminals), SDMA, FDMA TDMA, CDMA, Wireless LAN/(IEEE 802.11) Mobile Network Layer IP end Mobile IP Network Layers, Packet Delivery and Handover Management, Location Management, Registration. Tunneling and Encapsulation, Route Optimization, DHCP

Mobile Transport Layer: Conventional TCP/IP Protocols, Indirect TCP, Snooping TCP, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Database Issues: Database Hoarding & Caching Techniques. Client-Server Computing a Adaptation, Transnational Models, Query processing, Data Recovery Process & QoS Issues.

Data Dissemination and Synchronization: Communications Asymmetry. Classification of Data Delivery Mechanisms. Data Dissemination, Broadcast Models. Selective Tuning and Indexing Methods. Data Synchronization – Introduction. Software, and Protocols.

Mobile Ad hoc Networks (MANETs): Introduction, Applications a Challenges of a MANET Routing, Classification of Routing Algorithms. Algorithms such as DSR. AODV. DSDV. etc. , Mobile Agents. Service Discovery.

Text Books:

1. Joctien Schuller, "Mobile Communication". Addison-Wesley. Second Edition, 2009.
2. Raj Kamal. "Moblie Computing". Oxford University Press, 2007. ISBN: 0195686772

Reference Books:

1. Jochen Sdiiler. "Mobile Communications". Addison-Wesley. Second Edition. 2004.
2. Stomenovlc and Cacute, Handbook of Wireless Networks and Mobile Computing. Wiley 2002. ISBN 0471419028.
3. Reza Behravanfar. Mobile Computing Principles: Designing and Developing Mobile Applications with UML and XML. ISBN:0521817331. Cambridge University Press. 2004.

SOFT COMPUTING AND NEURAL NETWORKS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The student should be made to:

1. Learn the various soft computing frame works
2. Be familiar with design of various neural networks
3. Be exposed to fuzzy logic
4. Learn genetic programming.
5. Be exposed to hybrid systems.

Course Outcomes:

Upon completion of the course, the student should be able to:

1. Apply various soft computing frame works.
2. Design of various neural networks.
3. Use fuzzy logic.
4. Apply genetic programming.
5. Discuss hybrid soft computing.

SYLLABUS

Introduction: Artificial neural network: Introduction, characteristics- learning methods – taxonomy – Evolution of neural networks- basic models - important technologies - applications. Fuzzy logic: Introduction - crisp sets- fuzzy sets - crisp relations and fuzzy relations: cartesian product of relation - classical relation, fuzzy relations, tolerance and equivalence relations, non-iterative fuzzy sets. Genetic algorithm- Introduction - biological background - traditional optimization and search techniques - Genetic basic concepts.

Neural Networks: McCulloch-Pitts neuron - linear separability - hebb network - supervised learning network: perceptron networks - adaptive linear neuron, multiple adaptive linear neuron, BPN, RBF, TDNN- associative memory network: auto-associative memory network, hetero-associative memory network, BAM, hopfield networks, iterative autoassociative memory network & iterative associative memory network – unsupervised learning networks: Kohonen self organizing feature maps, LVQ – CP networks, ART network.

Fuzzy Logic: Membership functions: features, fuzzification, methods of membership value assignments- Defuzzification: lambda cuts - methods - fuzzy arithmetic and fuzzy measures: fuzzy arithmetic - extension principle - fuzzy measures - measures of fuzziness -fuzzy integrals - fuzzy rule base and approximate reasoning : truth values and tables, fuzzy propositions, formation of rules-decomposition of rules, aggregation of fuzzy rules, fuzzy reasoning-fuzzy inference systems-overview of fuzzy expert system-fuzzy decision making.

Genetic Algorithm: Genetic algorithm and search space - general genetic algorithm – operators - Generational cycle - stopping condition – constraints - classification - genetic programming – multilevel optimization – real life problem- advances in GA

Hybrid Soft Computing Techniques & Applications: Neuro-fuzzy hybrid systems - genetic neuro hybrid systems - genetic fuzzy hybrid and fuzzy genetic hybrid systems - simplified fuzzy ARTMAP - Applications: A fusion approach of multispectral images with SAR, optimization of traveling salesman problem using genetic algorithm approach, soft computing based hybrid fuzzy controllers.

Text Books:

1. J.S.R.Jang, C.T. Sun and E.Mizutani, “Neuro-Fuzzy and Soft Computing”, PHI / Pearson Education 2004.
2. S.N.Sivanandam and S.N.Deepa, "Principles of Soft Computing", Wiley India Pvt Ltd, 2011.

Reference Books:

1. S.Rajasekaran and G.A.Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis & Applications", Prentice-Hall of India Pvt. Ltd., 2006.
2. George J. Klir, Ute St. Clair, Bo Yuan, “Fuzzy Set Theory: Foundations and Applications” PrenticeHall,1997.
3. David E. Goldberg, “Genetic Algorithm in Search Optimization and Machine Learning” PearsonEducationIndia,2013.
4. James A. Freeman, David M. Skapura, “Neural Networks Algorithms, Applications, and ProgrammingTechniques,PearsonEducationIndia,1991.
5. Simon Haykin, “Neural Networks Comprehensive Foundation” Second Edition, Pearson Education, 2005.

SOFTWARE ENGINEERING AND MINI PROJECT LAB**Lab : 3 Periods****Sessionals : 50****Exam : 3 Hrs.****Ext. Marks : 50****Credits : 2****Course Objectives:**

The aim of this course is,

1. To design and development complex software solutions using the state of art software engineering techniques.
2. To work knowledge of UML, source control and project management.
3. To have deep knowledge of the technologies they used for implementing their project.
4. And know how to test and document software.

Course Outcomes :

1. Students will be **Construct, Design and implement** complex software solutions. [K3, K4].
2. Students will be able to test and document the software. [K3].
3. Students will be capable of working as part of a software team and develop significant projects under a tight deadline. [K3].
4. Students will be able **apply** the deep knowledge of the technologies they used for implementing their project. [K2].
5. Students will be able to **assess** the changes required for customization in project management. [K3]

SYLLABUS

The purpose of the Software Engineering Lab course is to familiarize the students with modern software engineering methods and tools, Rational Products. The course is realized as a project-like assignment that can, in principle, by a team of three/four students working full time. Typically the assignments have been completed during the semester by each project team.

The goal of the Software Engineering Project is to have a walk through from the requirements, design to implementing and testing. An emphasis is put on proper documentation. Term projects are projects that a group student might take through from initial specification to implementation by giving equal importance to both design and implementation.

Cycle I: Practicing UML diagrams using IBM Rational Rose. 6*3 periods= 18periods

Before developing a mini-project, in this cycle, the student is acquainted with different UML diagrams using Rational Rose. The experiments should include drawing UML diagrams listed below for two demo/example applications assigned by the lab Instructor. The input for the following experiments is problem statement for any two demo projects supplied by the instructor.

1. Introduction to Rational Rose and Practicing the following diagrams

- a. Activity diagrams for the overall business process of the projects
- b. Use-case diagram for the demo projects along with Use-case descriptions and sub-diagrams for Use-cases.

2. Class diagram- Class diagrams including the features like classes, relationships, attributes and methods along with their visibilities.

3. Interaction diagrams- Sequence diagrams and Collaboration diagrams for different scenarios of the systems with all features like actors, objects and interactions.

4. Activity diagrams, State chart and other diagrams - Activity diagrams including the features like fork join and swim lanes. State diagrams including composite states and transitions. Component diagrams, Package diagrams and Deployment diagrams.

5. Forward and Reverser Engineering- Forward Engineering Class diagrams to classes in C++ and java and persistent classes to a database. Reverse Engineering C++ code, java code and a database.

6. Documentation using Rational Rose clear quest.

Cycle II: Mini-Project 8*3 periods= 24periods

The project deliverables include

- Problem statement
- Requirements Analysis
- Design

A Software Design Description and a System Design.

A test specification.

- Implementation

Implement the assigned project with one of the following web technologies

- Front end: Java technologies/PHP/MS.NET Technologies
- Backend: Oracle/My-SQL/SQL-Server
- Testing

Reference Books :

1. Project-based software engineering: An Object-oriented approach, Evelyn Stiller, Cathie LeBlanc, Pearson Education.
2. Visual Modeling with Rational Rose 2002 and UML, Terry Quatrini, Pearson Education.

COMPUTER GRAPHICS & MULTIMEDIA LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objective:**

The computer graphics and multimedia laboratory is established for the purpose of providing working area for development of computer graphics and multimedia

Course Outcome

Create and evaluate graphic design projects using computer graphics software

LIST OF EXPERIMENTS

1. To implement Bresenham's algorithms for line, circle and ellipse drawing
2. To perform 2D Transformations such as translation, rotation, scaling, Reflection and shearing.
3. To implement Cohen–Sutherland 2D clipping and window–viewport mapping
4. To perform 3D Transformations such as translation, rotation and scaling.
5. User Interface Design & Graphics II: Create a user interface for your final project. Include 2 backgrounds and 1 button set. Aim for a cohesive look.
6. Multimedia Sound: Create 2 soundtracks and 2 EFX sounds for a previous project.
7. Procedure to create an animation to indicate a ball bouncing on steps
8. Procedure to simulate movement of a cloud.
9. Procedure to create an animation with the following features. WELCOME Letters should appear one by one The fill color of the text should change to a different color after the display of the full word.
10. Procedure to create an animation to represent the growing moon
11. Procedure to extract the flower only from given photographic image and organize it on a background. Selecting your own background for organization.
12. Procedure to use appropriate tool(s) from the toolbox cut the objects from 3 files (f1.jpg, f2.jpg & f3.jpg); organize them in a single file and apply feather effects.

Reference Books :

1. Vaughan, T. "Multimedia – Making it work (5th edition) ", McGraw-Hill.
2. Boyle, T. "Design for Multimedia Learning", Prentice-Hall, 1997.

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and Pu
3. shpaLatha, Second Edition, OUP.
4. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods,Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V:Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

COMPETITIVE CODING
(Common to CSE & IT)
MOOCS-III

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

1. To understand the essentials of competitive coding
2. To learn about competitive programming like time and space complexities
3. To investigate different methods like dynamic Arrays, Set & Map structures and sorting
4. To learn about String, Tree, Graph Theory algorithms

Course Outcomes:

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of competitive coding
2. Acquire Programming knowledge on time & space complexities
3. Acquire coding knowledge on dynamic Arrays, Set & Map structures and sorting
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on String, Tree, Graph Theory algorithms

SYLLABUS**UNIT-I Introduction to Competitive Coding**

Introduction to Competitive coding and coding Platforms. Coding solution Vs. Efficient Coding solution. Types of solution approaches. Analyzing problem specific data requirement, Various data representations. Essential Data structures for fast coding. Various Syntactical I/O techniques comparison. Numbers, operations (including exponentiation). Integer properties(positive, negative, even, odd, divisible, prime, etc), Fractions, percentages and ratios. Point, vector, Cartesian coordinates(2D integer grid).

UNIT-II Essentials to Competitive coding

Basic data structures: Arrays, Strings, Stacks, Queues, Linked Lists . Asymptotic Notations – (Big-O), Evaluating the runtime complexity – Space Complexity - Towers-of-Brahma – Standard Template Libraries - Square root functions, primality testing and related techniques. Euclidean algorithms. Recursion techniques. Organizing data in $O(n \log n)$. Binary search techniques. Red-Black Trees. Fenwick tree, Segment Tree.

Basic Techniques

Dynamic Arrays, Set structures, Map structures, Iterators and ranges, Generating Subsets, Generating permutations, Backtracking techniques, Pruning the search. Bit masking. Disjoint set union.

UNIT-III Essential Coding Algorithms

Selection based algorithms: sorting, Coin change problem, Fractional selections, Schedules matching, Activity marking, heap sort, Huffman coding techniques, Spanning Trees, Minimizing sums, Data compression. Finding method count, Subsequence and related problems, paths in grid. DP with Bit mask

UNIT-IV String & Tree Algorithms

TRIE data structure, Naïve string searching, z-algorithm, Manacher's algorithm, Rabin-Karp, KMP Algorithm, Tree Traversals, Diameter, All longest paths, Binary trees, Applying search property to tree structures. Suffix arrays.

UNIT-V Graph Algorithms

Graph Algorithms – DFS, BFS. Depth First and Breadth First Traversals - Shortest paths: Dijkstra's algorithm Bellman-Ford Algorithm – Floyd Warshall Algorithm - Adjacency List Representation – Euler path, tour, cycle – Eulerian Graph - Johnson's Algorithm for All-pairs shortest path – Shortest path in Directed Acyclic Graph. Bridges and articulation points. Topological sorting, Strongly connected components in directed graphs. 2-SAT.

Note: Introduce C++ STL/Java Collections and let students solve problems using C++ STL/Java Collections

References:

1. Data Structures and Algorithm Analysis in C – Mark Allen Weiss, Pearson Edu Publishers.
2. Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, Tata Mc Graw Hill Publishers.
3. Advanced Data Structures – Peter Brass, Cambridge University Press, 2008.
4. Introduction to Algorithms by Thomas H. Cormen, 3rd Edition, PHI.
5. <https://www.geeksforgeeks.org/>.
6. <https://www.tutorialspoint.com/>

**AMAZON WEBSERVICES
(MOOCS-III)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course Objective:

Amazon Web Services (AWS) is a comprehensive, evolving cloud computing platform provided by Amazon. It provides a mix of infrastructure as a service (IaaS), platform as a service (PaaS) and packaged software as a service (SaaS) offerings. It Provides Virtual servers, and also provides scalable object storage for data backup.

Course Outcome:

By the end of the course, student will be able to deploy their projects into cloud and they develop their projects by using AWS.

SYLLABUS**UNIT-I:**

1. Creating AWS account
2. Free tier Eligible services
3. Understanding AWS Regions and availability zones

UNIT-II:

EC2 (Elastic Cloud Comput)

- About EC2 and types , Pricing
- EIP (Elastic IP address), Allocating, associating , releasing
- Launch windows and Linux Instances in AWS
- Connecting windows and Linux instances from windows desktop and Linux machines

UNIT-III:

S3 (Simple Storage Service)

- About AWS Storage services, EBS and S3
- Creating S3 Buckets and putting objects in bucket
- Discussion about Bucket Properties
- S3 Pricing
- About S3 glacier

UNIT-IV:

EBS (Elastic Block Storage)

- Types of EBS Volumes
- Creation, attaching and Detaching volumes

UNIT-V:

ELB (Elastic Load Balancer)

- Understanding the load balancing
- Configuring ELB and adding the webserver under ELB

Auto Scaling

- Types of Scaling (Horizontal and Vertical)
- Configuring Launch Configuration
- Creating and defining the auto scaling group policy

UNIT-VI:

IAM (Identity Access Management)

- Understanding of AWS Security using IAM
- Definition of Roles, policies and Groups
- Creating IAM Users and managing password policies

RDS (Relational Database server)

- About RDS and available RDS Engines in AWS
- Configuring MYSQL RDS service
- Connecting EC2 Instance to RDS Instance

WEB LINKS:

<https://www.udemy.com/aws-certified-developer-associate/?>

**ASP.NET
(MOOCS-III)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course Objectives:

1. Introduction to web technologies.
2. Web Forms Architecture.
3. ASP.Net and HTTP.
4. Web application developing using Visual Studio.
5. State Management and Web Applications.
6. ASP.Net Server-side controls.
7. Caching in ASP.Net.
8. ASP.Net application configuration.
9. Debugging, Diagnostics of application.
10. Personalization and Security

Course Outcomes:

1. To successfully build database-driven Web applications and Web Sites.
2. To build web-based enterprise applications using ASP.NET and Visual Studio.
3. It is easy to develop the Web Services using .Net framework in Service-oriented Architectures.

SYLLABUS**Chapter 1:****Introduction to ASP.NET**, From ASP to ASP.NET, Web Forms, Web Services ASP.NET Features**Chapter 2:****Web Forms Architecture**, Page Class, Web Forms Life Cycle, Web Forms Event Model, Code-Behind**Chapter 3:****ASP.NET and HTTP**, Request/Response Programming, HttpRequest Class, HTTP Collections, HttpResponse Class, Redirection, HttpUtility Class

Chapter 4:

Web Applications Using Visual Studio, Using Visual Web Developer, Visual Studio Forms Designer, Using Components, Shadow Copying, Using the Global.asax File, Data Binding

Chapter 5:

State Management and Web Applications, Session State, Application State, Multithreading Issues, Cookies

Chapter 6:

Server Controls, HTML Server Controls, Web Forms Server Controls, Rich Controls, Validation Controls, User Controls.

Chapter 7:

Caching in ASP.NET, What Is Caching, Page-Level Caching, Page Fragment Caching, Optimizing Your ASP.NET Application, Application Caching

Chapter 8:

ASP.NET Configuration and Security Fundamentals, Configuration Overview, Authentication and Authorization, Forms Authentication, Windows Authentication, Security and ASP.NET

WEB LINKS:

<https://www.asp.net/freecourses>

https://mva.microsoft.com/en-us/training-courses/introduction-to-aspnet-mvc-8322?l=nKZwZ8Zy_3504984382

ROUTING & SWITCHING (MOOCS-III)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objective :

CCNA R&S: Introduction to Networks (ITN) covers networking architecture, structure, and functions. The course introduces the principles and structure of IP addressing and the fundamentals of Ethernet concepts, media, and operations to provide a foundation for the curriculum.

Course Outcomes :

By the end of the course, students will be able to:

1. Explain network technologies.
2. Explain how devices access local and remote network resources.
3. Describe router hardware.
4. Explain how switching operates in a small to medium-sized business network.
5. Design an IP addressing scheme to provide network connectivity for a small to medium-sized business network.
6. Configure initial settings on a network device.
7. Implement basic network connectivity between devices.
8. Configure monitoring tools available for small to medium-sized business networks.

SYLLABUS

Chapter 1 - Explore the Network

Chapter 2- Configure a Network Operating System

Chapter 3 - Network Protocols and Communications

Chapter 4 - Network Access

Chapter 5 - Ethernet

Chapter 6 - Network Layer

Chapter 7 - IP Addressing

Chapter 8- Subnetting IP Networks

Chapter 9 - Transport Layer

Chapter 10 - Application Layer

Chapter 11 - Build a Small Network

WEB LINK:

<https://www.cisco.com/>

DATA SCIENCE USING ADVANCED PYTHON (MOOCS-IV)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

1. Introduction to Data Science
2. Exploring steps in Data Science

Course Outcomes:

1. To work on data easily.
2. Familiar with Various modules for exploring on data like processing , visualization and statistical data analysis.
3. Able to work on real time data

SYLLABUS

Chapter 1: Introduction to Data Science

Introduction, Data Science environment setup,Pandas: Dimension & Description, Series, DataFrame, Panel (Heterogeneous data, Size Mutable, Data Mutable), Numpy: Operations using NumPy, NumPy – A Replacement for MatLab, ndarray Object, SciPy: SciPy Sub-packages, Data Structure, Matplotlib.

Chapter 2: Data Processing

Data Operations: Data Operations in Numpy, Data Operations in Pandas, Pandas Series, Pandas DataFrame, Pandas Panel, **Data Cleansing:** Analysis of missing data, Cleaning / Filling Missing Data, Replace Missing values, Processing CSV Data, Relational Databases, **Data Wrangling:** Merging Data, Grouping Data, Concatenating Data, **Data Aggregation:** Applying Aggregations on DataFrame, Processing Unstructured Data, Word Tokenization, Stemming and Lemmatization

Chapter 3: Data Visualization

Chart Properties, Chart Styling, Box Plots, Heat Maps, Scatter Plots, Bubble Charts, 3D Charts, Time Series, Geographical Data, Graph Data

Chapter 4: Statistical Data Analysis

Measuring Central Tendency, Measuring Variance, Normal Distribution, Binomial Distribution, Poisson Distribution, Bernoulli Distribution, P-Value, Correlation, Chi-Square Test, Linear Regression

Chapter 5: Project Development**WEB LINKS:**

<https://www.coursera.org/specializations/data-science-python>

**ANGULAR JS
(MOOCS-IV)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course objectives:

1. Reduce the amount of code you write to build rich user interface applications.
2. Increase the reliability and maintainability of UI by using data binding.
3. Retrieve data from back-end server, manipulate it and display it with ease.
4. Modularise your code with the custom services and directives.
5. Providing two way binding of data.
6. Create Single Page Applications (SPA).

Course Outcomes:

1. The main objective of AngularJS is to reduce the code to build user interface application.
2. To create single page applications.
3. To restore data from back-end server and manipulate it easily

Module 1: Introduction

Introduction to AngularJS, MVC Architecture, Conceptual Overview, Setting up the Environment, First Application, Understanding ng attributes.

Module 2: Expressions and Data Binding

Number and String Expressions , Object Binding and Expressions, Working with Arrays, Forgiving Behavior, Understanding Data binding.

Module 3: Working with Directives

Conditional Directives ,Styles Directives, Mouse and Keyboard Events Directives.

Module 4: Controllers

Understanding Controllers , Programming Controllers & \$scope object, Adding Behavior to a Scope Object, Passing Parameters to the Methods, Having Array as members in Controller Scope. Nested Controllers and Scope Inheritance, Multiple Controllers and their scopes.

Module 5: Filters

Built-In Filters , Uppercase and Lowercase Filters, Currency and Number Formatting Filters, OrderBy Filter. Filter Filter ,Creating Custom Filter.

Module 6: Forms

Using Simple Form, Working with Select and Options, Input Validations, Using CSS classes, Form Events, Custom Model update triggers, Custom Validations.

Module 7: Modules

Why Module? , Module Loading and Dependencies, Recommended Setup of Application, Creation vs Retrieval.

Module 8: Services

Understanding Services, Developing Creating Services, Using a Service, Injecting Dependencies in a Service.

Module 9: Ajax in AngularJS

\$http Service, \$q Service, Ajax Impl using \$http and \$q Service.

Module 10: Routing

Introduction to SPA , Creating HTML Templates, Configuring Route Provider.

WEB LINKS:

<https://www.udemy.com/angularjs/>

<https://www.coursera.org/learn/single-page-web-apps-with-angularjs>

C#.NET and VB.NET (MOOCS-IV)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

1. Learn the fundamentals of C# programming in Visual Studio.
2. Working with variables, data types.
3. Exception Handling in C#.
4. Object oriented programming in C#.
5. Use anonymous types, lambda expressions, extension methods, object initializers, and implicit type declarations.
6. To build desktop applications using Windows Forms in VB.NET.
7. To interact with relational databases using ADO.NET

Course Outcomes:

1. Understand .NET Framework and describe some of the major enhancements to the new version of Visual Basic.
2. Describe the basic structure of a Visual Basic.NET project and use main features of the integrated development environment (IDE).
3. Create applications using Microsoft Windows Forms Create applications that use ADO. NET

SYLLABUS

Chapter 1:C# Overview, First C# Console Application, Namespaces, Data Types, Conversions, Control Structures, Subroutines and Functions, Parameter Passing, Strings, Arrays, Console I/O, Formatting, Exception Handling

Chapter 2 : Object-Oriented Programming in C#, Classes, Access Control, Methods and Properties, Asymmetric Accessor Accessibility, Static Data and Methods, Inheritance, Overriding Methods, Abstract Classes, Sealed Classes, Access Control and Assemblies, Auto-Implemented Properties, Implicitly Typed Variables, Object Initializers, Collection Initializers, Anonymous Types, Partial Methods, Extension Methods, Lambda Expressions, Language-Integrated Query (LINQ)

Chapter 3: VB .NET - Features,IDE- Menu System, Toolbars,Code Designer,Solution Explorer, Object Browser, Toolbox, Class View Window, Properties Window,Server Explorer, Task List,Output Window, Command Window

.

Chapter 4: Elements of VB.net - Properties, Events and Methods of Form, Label, TextBox, ListBox, Combo Box, RadioButton, Button, Check Box, Progress Bar, Date Time Picker, Calendar, Picture Box, HScrollbar, VScrollbar, Group Box, ToolTip, Timer.

Chapter 5: Programming in VB.net: Data Types, Keywords, Declaring Variables and Constants, Operators, Understanding Scope and accessibility of variables, Conditional Statements- If- Then, If-Then-Else, Nested If, Select Case, Looping Statement- Do loop, For Loop, For Each-Next Loop, While Loop, Arrays- Static and Dynamic.

Chapter 6: Functions, Built-In Dialog Boxes, Menus and Toolbar

Menus and toolbars- Menu Strip, Tool Strip, Status Strip, Built-In Dialog Boxes –Open File Dialogs, Save File Dialogs, Font Dialogs, Color Dialogs, Print Dialogs, InputBox, Msg Box, Interfacing With End user- Creating MDI Parent and Child, Functions and Procedures- Built-In Functions- Mathematical and String Functions, User Defined Functions and Procedures.

Chapter 7: Advanced Concepts in VB.Net

Object Oriented Programming- Creating Classes , Objects, Fields, Properties, Methods, Events , Constructors and destructors, Exception Handling- Models, Statements, File Handling- Using File Stream Class, File Mode, File Share, FileAccess Enumerations, Opening or Creating Files with File Stream Class, Reading and Writing Text using StreamReader and StreamWriter Classes, Data Access with ADO.Net – Data Access with Server Explorer, Data Adapter and DataSets, ADO.NET Objects and Basic SQL.

Text Books:

1. Visual Basic.Net Black Book by Steven Holzner Dreamtech Press
2. The Complete Reference Visual Basic .NET Jeffery R. Shapiro Tata McGraw Hills Reference Books: Murach's Beginning Visual basic .Net By Anne Bohem

WEB LINKS:

<https://www.coursera.org/learn/introduction-programming-unity>

<https://www.pluralsight.com/courses/dotnet-csharp-tutorial>

<https://www.udemy.com/learn-vbnet/>

MATLAB (MOOCS-IV)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

1. Understanding the MATLAB environment.
2. Being able to do simple calculations using MATLAB.
3. Being able to carry out simple numerical computations and analyses using MATLAB

Course Outcomes:

1. Understand the main features of the MATLAB development environment.
2. Use the MATLAB GUI effectively.
3. Design simple algorithms to solve problems.
4. Write simple programs in MATLAB to solve scientific and mathematical problems

SYLLABUS

Chapter 1: Introduction, Basic features, A minimum MATLAB session, Starting MATLAB, Using MATLAB as a calculator, Quitting MATLAB, Creating MATLAB variables, Overwriting variable, Error messages, Making corrections, Controlling the hierarchy of operations or precedence, Controlling the appearance of floating point number, Managing the workspace, Keeping track of your work session, Entering multiple statements per line, Miscellaneous commands

Chapter 2: Mathematical functions, Basic plotting: overview, Creating simple plots, Adding titles, axis labels, and annotations, Multiple data sets in one plot, Specifying line styles and colors, Matrix generation: Entering a vector, Entering a matrix, Matrix indexing, Colon operator, Linear spacing, Colon operator in a matrix, Creating a sub-matrix, Deleting row or column, Dimension, Continuation, Transposing a matrix, Concatenating matrices, Matrix generators, Special matrices

Chapter 3: Array operations and Linear equations, Array operations: Matrix arithmetic operations, Array arithmetic operations, Solving linear equations: Matrix inverse, Matrix functions

Chapter 4: Introduction to programming in MATLAB, Introduction, M-File Scripts: Examples, Script side-effects, M-File functions: Anatomy of a M-File function, Input and output arguments, Input to a script file, Output commands

Chapter 5: Control flow and operators, The “if...end” structure, Relational and logical operators, The “for...end” loop, The “while...end” loop, Other flow structures, Operator precedence, Saving output to a file

Chapter 6: Debugging M-files, Debugging process: Preparing for debugging, Setting breakpoints, Running with breakpoints, Examining values, Correcting and ending debugging, Ending debugging, Correcting an M-file

WEB LINKS:

<https://www.edx.org/learn/matlab>

<https://matlabacademy.mathworks.com/>



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

INFORMATION TECHNOLOGY

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 4101	Cloud Computing	4	3	1	-	4	30	70	100
B16 IT 4102	Big Data Analytics	4	3	1	-	4	30	70	100
B16 ENG 4101	Principles of Economics And Management	4	3	1	-	4	30	70	100
B16 IT 4103	Knowledge Engineering Lab	2	-	-	3	3	50	50	100
B16 IT 4104	Network Programming Lab	2	-	-	3	3	50	50	100
B16 IT 4105	Project Phase-I	2	--	--	3	3	50	--	50
Total		18	9	3	9	21	240	310	550

CLOUD COMPUTING

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. The student will learn about the basic of cloud environment, building software systems and components that scale to millions of users in modern internet cloud concepts capabilities.
2. The student will learn about the various cloud service models including IaaS, PaaS, SaaS.
3. The student will learn about developing cloud based software applications on top of cloud platforms.
4. The student will learn about the cloud governance and Security issues in Cloud Environment.

Course Outcomes:

1. The student will understand the cloud environment.
2. The student will understand and learn the various Cloud based Services.
3. The student will able to develop cloud based applications.
4. The Student understands the security, governance and Economic in Cloud computing.

SYLLABUS

Introduction to cloud computing: Cloud computing components, Infrastructure services, storage applications, database services – introduction to SaaS, PaaS, IaaS, IdaaS, data storage in cloud, Virtualization: enabling technologies, types of virtualization, server virtualization, desktop virtualization, memory virtualization, application and storage virtualization, tools and products available for virtualization.

SAAS and PAAS: Getting started with SaaS, Software plus Services - Overview SaaS solutions, SOA, PaaS and benefits.

IaaS and Cloud Data Storage: understanding IaaS, improving performance for load balancing, server types within IaaS, utilizing cloud based NAS devices, cloud based data storage, and backup services, cloud based block storage and database services.

Cloud Application development: Client server distributed architecture for cloud designing cloud based solutions, coding cloud based applications, traditional Apps vs cloud Apps, fundamental treatment of web application frameworks. Web Applications, Web APIs, Web Browsers.

Cloud Governance and economics: Securing the cloud, disaster recovery and business continuity in the cloud, Managing the cloud, migrating to the cloud, governing and evaluating the clouds business impact and economics, Inside Cloud: Introduction to MapReduce and Hadoop-over view of big data and its impact on cloud.

Text Books:

1. Cloud Computing: SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and More, Kris Jamsa, Jones & Bartlett Publishers, Paper back edition, 2013.
2. Cloud Computing: A Practical Approach, Anthony T .Velte, Toby J.Velte, Robert Elsenpeter, Tata McGraw Hill Edition

Reference Books:

1. Hadoop MapReduce cookbook, SrinathPerera and ThilinaGunarathne, Packet publishing

BIG DATA ANALYTICS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To explore the Bigdata analytics Objectives, Purpose, Scope.
2. To provide advanced knowledge and skills in the field of Big Data analytics.
3. To study the Building blocks of Hadoop and its Functionalities to handle Big Data.
4. To write mapreduce programming using Java to solve basic and advanced bigdata problems.
5. Capable to quickly adapt to new technology in the field of Big Data, assimilate new information, and solve real world problems.

Course Outcomes: End of this course student can able to

1. Understand Big Data and its characteristics.
2. Understand basic Building Blocks of Hadoop and its functionalities.
3. Understand how the big data is stored in HDFS and how Map Reduce processes this data stored in HDFS.
4. Design Map Reduce programs to handle basic and advanced problems by using Hadoop architecture.
5. Identify the challenges in Big Data with respect to IT Industry.

SYLLABUS

Introduction to Big Data & Hadoop: Data, Big Data-definition, Categories Of Big Data, Characteristics of Big Data (Volume, Variety, Velocity, Veracity, Validity), Importance of Big Data, Patterns for Big Data Development, Hadoop- definition, Understanding distributed systems and Hadoop, Comparing SQL databases and Hadoop, Understanding MapReduce, Data in the Warehouse and Data in Hadoop, Hadoop versions.

Introduction to Hadoop Architecture: Hadoop Components, Hadoop Echo system, Building blocks of Hadoop: NameNode, DataNode, Secondary NameNode, JobTracker and Task Tracker, Data Types In Hadoop, I/O in Hadoop, Analysing data with Hadoop, Map Reduce implementation architecture, Casestudy: Counting words with Hadoop.

MapReduce -A Weather Dataset, Analyzing the Data with Hadoop, Scaling Out, Hadoop Streaming, Hadoop Pipes, Developing a MapReduce Application - The Configuration API, Configuring the Development Environment, Running Locally on Test Data, Running on a Cluster, Tuning a Job: Map side tuning & Reduce side tuning.

HDFS: Anatomy of a MapReduce program, Reading and writing the Hadoop Distributed File system -The Design of HDFS, HDFS Concepts, The Command-Line Interface, Hadoop Filesystem, Data Flow, Parallel Copying with distcp, Hadoop Archives.

MapReduce Programming: Writing basic Map Reduce programs, Counting things, Adapting for Hadoop's API changes, Streaming in Hadoop, Improving performance with Combiners, Chaining MapReduce jobs, Partitioning into multiple output files. Casestudy: Patent data set.

MapReduce Advanced Programming: Advanced MapReduce, creating a Bloom filter, Passing job-specific parameters to your tasks, , Inputting from and outputting to a database. Modeling data and solving problems with graphs, Shortest Path Algorithm, Friends-of-Friends Algorithm, PageRank Algorithm.

Text Books:

1. Dirk deRoos, Chris Eaton, George Lapis, Paul Zikopoulos, Tom Deutsch
“Understanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data”, 1st Edition, TMH, 2012.
2. Hadoop: The Definitive Guide by Tom White, 3rd Edition, O’reilly

Reference Books:

1. Hadoop in Action by Chuck Lam, MANNING Publ.
2. Hadoop in Practice by Alex Holmes, MANNING Publishers
3. Mining of massive datasets, AnandRajaraman, Jeffrey D Ullman, Wiley Publications.

PRINCIPLES OF ECONOMICS & MANAGEMENT
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Apply economic reasoning to the analysis of selected contemporary economic problems.
2. Understand how households (demand) and businesses (supply) interact in various market structures to determine price and quantity of goods and services produced and consumed.
3. Analyze the efficiency and equity implications of government interference in markets.
4. Recognize and identify situations leading to market failures and government failures.

Course Outcomes:

1. Understand the links between production costs and the economic models of supply.
2. Represent supply, in graphical form, including the upward slope of the supply curve and what shifts the supply curve.
3. Understand the efficiency and equity implications of market interference, including government policy.
4. Understand how different degrees of competition in a market affect pricing and output.
5. Apply economic reasoning to individual and firm behavior.

SYLLABUS

Introduction to Managerial Economics: Wealth, Welfare and Scarce Definitions of Economics; micro and Macro Economics; Demand- Law of Demand, Elasticity of Demand, types of Elasticity and factors of determining price elasticity of Demand: utility- Law of Diminishing Marginal Utility and its limitations.

Conditions of Different Market Structures: Perfect Competition, Monopolistic Competition, Monopoly, Oligopoly, and Duopoly.

Forms of Business Organizations: Sole Proprietorship, Partnership, Joint Stock Company- Private Limited and Public Limited Companies, Public Enterprises and their types.

Introduction to Management: Functions of Management- Taylor's Scientific management; Henry Fayol's Principle of Management; Human Resource Management- basic Functions of HR Manager; Man Power Planning, Recruitment, Selection, Training, Development, Placement, Compensation and performance Appraisal(in brief).

Production Management: Production Planning and Control, plant Location, Break- Even Analysis, assumptions and applications.

Financial Management: Types of Capital: Fixed and Working Capital , and Methods of Raising Finance; Depreciation: Straight Line and Diminishing Balance Methods. Marketing Management: Functions of marketing and Distribution Channels.

Entrepreneurship: Entrepreneurial Functions, Entrepreneurial Development: Objectives, Training, Benefits: Phases of Installing a project

Text Books :

1. K.K.DEWETT, Modern Economic Theory, S.Chand and Company, New Delhi-55.
2. S.C. Sharma and Banga T. R., Industrial Organization & Engineering Economics, khanna Publications, Delhi-6.

Reference Books

1. A.R. AryaSri, Management Science, TMH publications, New Delhi-20.
2. A.R. AryaSri, Managerial Economics and Financial Analysis, TMH Publications, new Delhi-20.

KNOWLEDGE ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To study the various data analysis techniques in R Programming language.
2. To apply the various data mining techniques available in WEKA for generating Knowledge such as Association Analysis, Classification and Clustering to various standard datasets and own datasets.
3. To build Rule based Knowledge Systems using forward chaining and backward chaining using CLIPS and PROLOG respectively

Course Outcomes:

1. Student will be able to write R programs to perform several data analytics operations on datasets
2. Ability to extract patterns by applying appropriate data mining techniques from different types of datasets using WEKA.
3. Ability to apply knowledge represented in the form of rules to draw conclusions using either forward or backward chaining using CLIPS /PROLOG.

SYLLABUS

Exploratory data analysis using R:

1. Load the 'iris. CSV' file and display the names and type of each column. Find statistics such as min, max, range, mean, median, variance, standard deviation for each column of data.
2. Write R program to normalize the variables into 0 to 1 scale using min-max normalization.
3. Generate histograms for any one variable (sepal length/ sepal width/ petal length/ petal width) and generate scatter plots for every pair of variables showing each species in different color.
4. Generate box plots for each of the numerical attributes. Identify the attribute with the highest variance.
5. Study of homogeneous and heterogeneous data structures such as vector, matrix, array, list, data frame in R.
6. Write R Program using 'apply' group of functions to create and apply normalization function on each of the numeric variables/columns of iris dataset to transform them into a value around 0 with z-score normalization.
7. a) Use R to apply linear regression to predict evaporation coefficient in terms of air velocity using the data given below:

Air Velocity (cm/sec)	20,60,100,140,180,220,260,300,340,380
Evaporation Coefficient (sqmm/sec)	0.18, 0.37, 0.35, 0.78, 0.56, 0.75, 1.18, 1.36, 1.17, 1.65

- b) Analyze the significance of residual standard-error value, R-squared value, F-statistic. Find the correlation coefficient for this data and analyze the significance of the correlation value.

- c) Perform a log transformation on the 'Air Velocity' column, perform linear regression again, and analyze all the relevant values.

WEKA Knowledge Extraction toolkit:

8. Create an ARFF (Attribute-Relation File Format) file and read it in WEKA. Explore the purpose of each button under the preprocess panel after loading the ARFF file. Also, try to interpret using a different ARFF file, *weather.arff*, provided with WEKA.
9. Performing data preprocessing in Weka
Study **Unsupervised Attribute Filters** such as **Replace Missing Values** to replace missing values in the given dataset, **Add** to add the new attribute *Average*, **Discretize** to discretize the attributes into bins. Explore **Normalize** and **Standardize** options on a dataset with numerical attributes.
10. Classification using the WEKA toolkit
Demonstration of classification process using id3 algorithm on categorical dataset (weather).
Demonstration of classification process using naïve Bayes algorithm on categorical dataset ('vote').
Demonstration of classification process using Random Forest algorithm on datasets containing large number of attributes.
11. Classification using the WEKA toolkit – Part 2
Demonstration of classification process using J48 algorithm on mixed type of dataset after discretizing numeric attributes.
Perform cross-validation strategy with various fold levels. Compare the accuracy of the results.
12. Performing clustering in WEKA
Apply hierarchical clustering algorithm on numeric dataset and estimate cluster quality.
Apply DBSCAN algorithm on numeric dataset and estimate cluster quality.
13. Association rule analysis in WEKA
Demonstration of Association Rule Mining on supermarket dataset using Apriori Algorithm with different support and confidence thresholds.
Demonstration of Association Rule Mining on supermarket dataset using FP-Growth Algorithm with different support and confidence thresholds.

Building Knowledge based Inference Systems:

14. Implement AI problem solving through Rule based forward chaining inference using public domain software tool like CLIPS.
15. Implement AI problem solving through Rule based Backward chaining inference using PROLOG

Reference Books:

1. Practical data science with R, Nina Zumel and John Mount- Dreamtech Press.

NETWORK PROGRAMMING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To provide graduates with a good foundation in computer science and engineering fundamentals required to solve engineering problems that will facilitate them to find employment in industry and/or to pursue postgraduate studies with an appreciation for lifelong learning.
2. To provide graduates with analytical and problem solving skills to design algorithms, other hardware/software systems, and inculcate professional ethics, inter-personal skills to work in a multi-cultural team.
3. To facilitate graduates get familiarized with state of the art software/hardware tools, imbibing creativity and Innovation that would enable them to develop cutting-edge technologies of multi-disciplinary nature for societal development.

Course Outcomes:

1. Students will be able to design and create well known ports on a Local/Remote System.
2. Students will be able to design One-One and many-many chat application by socket connection and displaying what is written by one party to the other
3. Students will be able to design data retrieval from a Remote database.
4. Students will be able to design SMTP Mail Client: Gives the server name, send email to the recipient
5. using SMTP commands and POP Client for retrieve the mails.

SYLLABUS

1. Create to identify well known ports on a Local/Remote System.
2. Design and create One-to-One chat application by opening socket connection and displaying what is written by one party to the other.
3. Design and create Many-to-Many chat (Broad cast) Each client opens a socket connection to that chat server and writes to the socket. Whatever is written by one party can be seen by all other parties.
4. Design Data retrieval from a Remote database:
At the remote database a server listens for client connections. The server accepts SQL Queries from the client executes it on the database and sends the responses to the client.
5. Design Mail SMTP Client: Gives the server name, send email to the recipient using SMTP commands.
6. Design and create POP Client: Gives the server name, user name and password, retrieve the mails and allow manipulation of mailbox using POP commands.
7. Design Simulation of Telnet: Provide a user interface to contact well known ports so that client server interaction can be seen by the user.
8. Design Simple file transfer between two systems (without protocols):
By opening socket connection to our server on one system and sending a file from one system to another.
9. Design Download load an Image file using URL Commands.

Reference Book:

1. Java Network Programming-Orielly

PROJECT PHASE-I**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

IV/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

INFORMATION TECHNOLOGY
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 IT 4201	Embedded Systems	4	3	1	-	4	30	70	100
# ELE-III	ELECTIVE-III	4	3	1	-	4	30	70	100
B16 IT 4207	Internet Of Things Lab	2	-	-	3	3	50	50	100
B16 IT 4208	Project Phase-II	12	-	-	9	9	50	100	150
	Total	22	6	2	12	20	160	290	450

# ELE-III	B16 IT 4202	Artificial Intelligence
	B16 IT 4203	Information Retrieval
	B16 IT 4204	Advanced Operating Systems
	B16 IT 4205	Software Project Management
	B16 IT 4206	E-Commerce

EMBEDDED SYTEMS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To study the basics of embedded systems and its examples.
2. To study the 8051 Microcontroller architecture and its instruction set.
3. To discuss various software architectures in embedded systems.
4. To discuss Inter Task Communication procedures in RTOS and design issues of RTOS.
5. To study various embedded software development tools and debugging techniques.

Course Outcomes:

1. Student will be understand the basic architecture of 8051 micro controller.
2. ability to write ALP programs using 8051 instruction set.
3. Ability to understand the concepts related to RTOS and its Inter Task Communication methods.
4. Ability to understand various design issues of RTOS.
5. Understand about embedded software development tools.

SYLLABUS

Introduction to Embedded Systems: Examples, Typical Hardware, Memory, Microprocessors, Busses; Introduction to 8051 Microcontroller, Architecture, Instruction set, Programming.

Interrupts: Interrupt Basics, Shared-Data problem, Interrupt Latency.

Software Architectures: Round-Robin Architecture, Round-Robin with Interrupts Architecture, Function-Queue Scheduling Architecture, Real-Time Operating Systems Architecture, Selection of Architecture.

Real Time Operating System: Tasks and Task States, Tasks and Data, Semaphores and Shared Data, Semaphore Problems, Semaphore variants.

Inter Task Communication: Message Queues, Mailboxes, Pipes, Timer Functions, Events, and Memory Management, Interrupt Routines in RTOS Environment.

Design issues of RTOS: Principles, Encapsulation Semaphores and Queues, Hard Real-Time Scheduling Considerations, Saving Memory Space, Saving Power.

Embedded Software development Tools: Host and Target Machines, Linker/Locator for Embedded Software, Getting Embedded Software into the Target System.

Embedded Software Debugging Techniques: Testing on your Host Machine, Instruction Set Simulators, Laboratory Tools used for Debugging.

Introduction to the Internet of Things: History of IoT, IoT Architecture, M2M–Machine to Machine, Web of Things, IoT protocols, The Layering concepts, IoT Communication Pattern, IoT protocol Architecture.

Text Books:

1. The 8051 Microcontroller Architecture, Programming & Applications, Kenneth J. Ayala, Penram International.
2. An Embedded Software Primer, David E. Simon, Pearson Education , 2005.
3. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Marina Ruggieri&HomayounNikookar, River Publishers Series in Communications.

Reference Book:

1. Embedded Systems: Architecture , Programming and Design, Raj Kamal, Tata McGraw- Hill Education, 2008

ARTIFICIAL INTELLIGENCE (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To learn the difference between optimal reasoning vs human like reasoning
2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. To learn different knowledge representation techniques
4. To understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

Course Outcomes:

1. Possess the ability to formulate an efficient problem space for a problem expressed in English.
2. Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
3. Possess the skill for representing knowledge using the appropriate technique.
4. Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing

SYLLABUS

Introduction to Artificial Intelligence, Artificial Intelligence Technique, Representation of a problem as State space search, production systems, Problem characteristics, Production System characteristics.

Heuristic Search Technologies: Generate & Test Hill Climbing, Best First search, Problem reduction, Constraint satisfaction, MeansEnd Analysis.

Predicate Logic Proof with Backward Chaining, Resolution, question answering.

Representing Knowledge Using Rules: Procedural Vs Declarative knowledge, Logic Programming, Forward Vs Backward Reasoning, Matching, Control Knowledge.

Symbolic Reasoning with uncertainty: Non-monotonic Reasoning, Dependency – Directed Backtracking TMS. Statistical Reasoning with Bayes Theorem, certainty Factors & Rule Based System, DS-Theory.

Weak & Strong Slot Filler Structures: Semantic nets, Frames, Conceptual dependencies, Scripts.

Planning: Block world, Components of a Planning System, Goal State Planning, Non LinearPlanning, Hierarchical Planning.

Natural Language Processing and Expert Systems:Syntactic Analysis, Semantic Analysis, Discusses and Pragmatic Processing. Representing and Using Domain Knowledge, Expert Systems Shells, Explanation

Text Books:

1. Artificial Intelligence, Rich E & Knight K – Tata Mcgrahill (1991).
2. Introduction to Artificial Intelligence & Expert Systems, Paterson. PHI

Reference Books:

1. Artificial Intelligence – A modern approach, Stuart Russel, Peter Norwig, Pearosn Education.

INFORMATION RETRIEVAL (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To provide the foundation knowledge in information retrieval.
2. To equip students with sound skills to solve computational search problems.
3. To appreciate the different applications of information retrieval techniques in the Internet or Web environment.
4. To provide hands-on experience in building search engines and/or hands-on experience in evaluating search engines.

Course Outcomes:

1. Identify basic theories in information retrieval systems
2. Identify the analysis tools as they apply to information retrieval systems
3. Understands the problems solved in current IR systems
4. Understand the difficulty of representing and retrieving documents.
5. Explain the concepts of indexing, vocabulary, normalization and dictionary in information retrieval.
6. Implement retrieval systems for web search tasks.

SYLLABUS

Introduction to Information Storage and Retrieval System: structures, AlgorithmsIntroduction Domain Analysis of IR systems and other types of Information Systems, IRSystem Evaluation. Introduction to Data Structures and Algorithms related to InformationRetrieval: Basic Concepts, Data.

Inverted files: Introduction, Structures used in Inverted Files, Building Inverted file using assorted array, Modifications to Basic Techniques.

Signature Files: Introduction, Concepts of Signature Files, Compression, Vertical Partitioning, Horizontal Partitioning.

New Indices for Text: PAT Trees and PAT Arrays: Introduction, PAT Tree structure,Algorithms on the PAT Trees, Building PAT trees as PATRICA Trees, PAT representation asarrays.

Stemming Algorithms: Introduction, Types of Stemming Algorithms, Experimental Evaluations of Stemming to Compress Inverted Files

Thesaurus Construction: Introduction, Features of Thesauri, Thesaurus Construction, Thesaurus construction from Texts, Merging existing Thesauri

Text Books:

1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms, Prentice Hall, 1992.
2. Modern Information Retrieval By Yates Pearson Education.
3. Information Storage & Retrieval By Robert Korfhage – John Wiley & Sons.

Reference Books:

1. Kowalski, Gerald, Mark T Maybury: Information Retrieval Systems: Theory and Implementation, Kluwer Academic Press, 1997.
2. Information retrieval Algorithms and Heuristics, 2ed, Springer.

ADVANCED OPERATING SYSTEMS
(Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Following are the objectives of the course.

1. To understand distributed systems and their design issues and topologies.
2. To understand communication in client – server model and ATM networks.
3. To understand remote execution of procedures and associated issues.
4. To understand different protocols to achieve clock synchronization and mutual exclusion in distributed systems.
5. To understand organization of multiple processes in distributed systems and design and implementation issues of processor allocation.
6. To understand the design and implementation issues of distributed file system.
7. To understand distributed shared memory architectures and systems.

Course Outcomes:

1. Students will be able to understand distributed systems and hardware and software concepts and their combinations.
2. Students will be able to understand communication in ATM networks and Client – Server architecture.
3. Students will be able to understand principles of remote procedure execution and address related issues and discuss related problems.
4. Students will be able to understand clock synchronization and mutual exclusion in distributed systems and relevant protocols.
5. Students will be able to understand multiple processors organization and their allocation.

SYLLABUS

Introduction to Distributed Systems: Goals – Advantages of distributed systems over centralized systems – disadvantages of distributed systems, Hardware & Software Concepts, loosely coupled systems, Network operating systems, True Distributed System, Multiprocessor time sharing system. Design Issues: Transparency – Flexibility – Performance – Scalability--Reliability. ATM networks.

The Client – Server Model: Clients and Servers, Example, Addressing, Blocking, Buffering, Reliability, Implementing the Client-Server model. Remote Procedure Calling: Introduction, Basic Operation, Parameter passing, Dynamic binding, RPC semantics in the presence of failures, Implementation issues, RPC problem areas.

Synchronization in Distributed systems: Clock synchronization, Logical Clocks, Physical Clocks, Clock synchronization algorithms, Mutual exclusion: Centralized algorithm, Distributed algorithm, Token ring algorithm, comparison of the three algorithms, Election algorithms: The Bully algorithm, Ring algorithm, Dead Locks in distributed systems, Distributed deadlock detection.

Process and Processors in distributed systems: Threads, Introduction, Usage, Design issues for thread packages, System models, The workstation model, The processor pool model, The hybrid model, Processor allocation, Allocation models, Design issues, Implementation issues.

Distributed File Systems: Introduction, Distributed file system design, Distributed file system implementation, Sun's network file system.

Distributed shared memory: Introduction, Bus based multiprocessors, Ring based multiprocessors, Switched multiprocessors, Consistency models, Page based Distributed shared memory, Shared variable Distributed shared memory, Object based Distributed Shared Memory.

Text Book:

1. Distributed Operating systems, Andrew S. Tanenbaum, Pearson Education Publishers.

Reference Book:

1. Advanced Concepts in Operating Systems, Singhal and Niranjana G. Shivaratna

SOFTWARE PROJECT MANAGEMENT (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study how to plan and manage projects at each stage of the software development life cycle (SDLC)
2. To train software project managers and other individuals involved in software project planning and tracking and oversight in the implementation of the software project management process.
3. To understand successful software projects that support organization's strategic goals

Course Outcomes

1. To match organizational needs to the most effective software development model
2. To understand the basic concepts and issues of software project management
3. To effectively Planning the software projects
4. To implement the project plans through managing people, communications and change
5. To select and employ mechanisms for tracking the software projects
6. To conduct activities necessary to successfully complete and close the Software projects
7. To develop the skills for tracking and controlling software deliverables
8. To create project plans that address real-world management challenges

SYLLABUS

Introduction Project, Management, Software Project Management activities, Challenges in software projects, Stakeholders, Objectives & goals Project Planning: Step-wise planning, Project Scope, Project Products & deliverables, Project activities, Effort estimation, Infrastructure.

Project Approach Lifecycle models, Choosing Technology, Prototyping Iterative & incremental Process Framework: Lifecycle phases, Process Artifacts, Process workflows (Book 2)

Effort estimation & activity Planning Estimation techniques, Function Point analysis, SLOC, COCOMO, Usecase-based estimation , Activity Identification Approaches, Network planning models, Critical path analysis

Risk Management Risk categories, Identification, Assessment, Planning and management, PERT technique, Monte Carlo approach

Project Monitoring & Control , Resource Allocation Creating a framework for monitoring & control, Progress monitoring, Cost monitoring, Earned value Analysis, Defects Tracking, Issues Tracking, Status reports, Types of Resources, Identifying resource requirements, Resource scheduling

Software Quality Planning Quality, Defining Quality – ISO 9016, Quality Measures, Quantitative Quality Management Planning, Product Quality & Process Quality Metrics, Statistical Process Control Capability Maturity Model, Enhancing software Quality (Book3)

Text Books

1. Software Project Management, Bob Hughes & Mike Cotterell, TATA Mcgraw-Hill
2. Software Project Management, Walker Royce: Pearson Education, 2005.
3. Software Project Management in practice, Pankaj Jalote, Pearson.

Reference Book

1. Software Project Management, Joel Henry, Pearson Education.

E-COMMERCE (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives :

1. This course introduces the concepts, vocabulary, and procedures associated with E-Commerce and the Internet.
2. To explain the students an overview of all aspects of E-Commerce.
3. To explain the Topics include development of the Internet and E-Commerce, options available for doing business on the Internet, features of Web sites and the tools used to build an E-Commerce web site, marketing issues, payment options, security issues, and customer service.

Course outcomes:

1. Ability to discuss the e-Commerce process. Describe an example of system architecture for an e-Business. List the seven major elements of web design.
2. Ability to Identify and explain fundamental web site tools including design tools, programming tools, and data processing tools. Identify the major electronic payment issues and options.
3. Ability to discuss security issues and explain procedures used to protect against security threats.
4. Ability to Identify and discuss management issues underlying e-Commerce issues including organizational structure, strategic planning, goal setting, corporate social responsibility, changing market intermediaries, resource allocation and customer service.

SYLLABUS

Electronic commerce environment and opportunities: Back ground–The Electronic commerce Environment–Electronic Market Place Technologies. Modes of electronic commerce: Overview–EDI–Migration to open EDI–Ecommerce with WWW/Internet– Commerce Net Advocacy–Web commerce going forward.

Approaches to safe electronic Commerce: Overview– Source–Transport Protocols–Secure Transactions– Secure Electronic Payment Protocol–Secure Electronic Transaction–Certificates for Authentication–Security on Web Servers and enterprise networks.

Electronic cash and electronic payment schemes: Internet Monetary Payment and Security requirements–payment and purchase order process–online electronic cash.

Master card/ Visa Secure electronic transaction: Introduction – Business requirements - Concepts - Payment Processing. Email and Secure Email Technologies for Electronic Commerce:

Introduction –The means of Distribution –A model for Message Handling –How Does an Email Work.

Internet Resources for Commerce: Introduction –Technologies for Web Servers –Internet Applications for commerce – Internet Charges –Internet Access and Architecture–Searching the Internet.

Text Books:

1. Web Commerce Technology Hand Book Daniel Minoli, Emma Minoli McGraw Hill

Reference Book:

1. Frontiers of Electronic Commerce Ravi Kalakotar, Andrew B. Whinston Addison-Wesley

INTERNET OF THINGS LAB

LAB : 3 Periods

Exam : 3 Hrs.

Sessionals : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To assess the vision and introduction of IoT.
2. To Understand IoT Market perspective.
3. To Implement Data and Knowledge Management and use of Devices in IoT Technology.
4. To Understand State of the Art - IoT Architecture.
5. To classify Real World IoT Design Constraints, Industrial Automation in IoT.

Course Outcomes:

1. Ability to interpret the vision of IoT from a global context.
2. Ability to determine the Market perspective of IoT.
3. Ability to Compare and Contrast the use of Devices, Gateways and Data Management in IoT.
4. Implement state of the art architecture in IoT.
5. Illustrate the application of IoT in Industrial Automation and identify Real World Design Constraints.

SYLLABUS

1. Study of Various Network Protocols Used In Iot.
2. Application of WifiInIot Systems.
3. Application of 6lowpan InIot Systems.
4. Application of Bluetooth InIot Systems.
5. Application of 802.15.4 Zigbee. In Iot Systems.
6. Design A Simple Iot System Comprising Sensors, Wireless Network Connection, Data Analytics.

Reference Books:

1. The Internet of Things: Connecting Objects, HakimaChaouchi (Editor), ISBN: 978-1-84821-140-7, 288 pages, June 2010, Wiley-ISTE
2. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Marina Ruggieri&HomayounNikookar, River Publishers Series in Communications.

PROJECT PHASE-II**LAB : 9 Periods**

Sessionals	: 50
Ext. Marks	: 100
Credits	: 12

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation. External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:
 1. Cover Page & Title Page .
 2. Bonafide Certificate
 3. Abstract.
 4. Table of Contents
 5. List of Tables
 6. List of Figures
 7. List of Symbols, Abbreviations and Nomenclature
 8. Chapters
 9. Appendices
 10. References

*The table and figures shall be introduced in the appropriate places.